$$\omega_\epsilon := \begin{cases} \omega \setminus \bar{B}_\epsilon(\eta_0) & \text{for } \eta_0 \in \omega, \\ \omega \cup \bar{B}_\epsilon(\eta_0) & \text{for } \eta \in \Omega \setminus \overline{\omega}. \end{cases} \tag{1}$$

$$-\partial_t \bar{p}_\epsilon - \Delta \bar{p}_\epsilon = -2y^{\bar{u}_\epsilon, f, \epsilon} \quad \text{in } \Omega \times (0, T] \tag{2.2}$$

$$\min_{u \in L_2(0,T;L_2(\Omega))} J(\epsilon, u, f) = J(\epsilon, \bar{u}_\epsilon, f). \tag{A.3}$$

$$\lim_{\epsilon \searrow 0} \frac{J(0, \hat{\chi}_{\omega_\epsilon} \bar{p}_\epsilon, f) - J(0, \hat{\chi}_\omega \bar{p}_\epsilon, f)}{|B_\epsilon(0)|} = \frac{1}{4\gamma} \int_0^T (\bar{p}^{f,\omega}(\eta_0, t))^2 \, dt. \tag{4a}$$

$$J(0, \hat{\chi}_{\omega_\epsilon} \bar{p}_\epsilon, f) = \int_{\Omega_T} y_\epsilon^2 + \frac{1}{4\gamma} \chi_{\omega_\epsilon} \bar{p}_\epsilon^2 \, dx \, dt, \tag{5}$$

$$\partial y_\epsilon - \Delta y_\epsilon = \frac{1}{2\gamma} \chi_{\omega_\epsilon} \bar{p}_\epsilon \quad \text{in } \Omega \times (0, T] \tag{6.6}$$

$$\int_0^1 \partial_y \mathcal{L}(\epsilon, s y_\epsilon + (1-s) y_0, \bar{q}_\epsilon; \bar{p}_\epsilon)(\psi) = 0 \text{ for all } \psi. \tag{A.7}$$

$$-\partial_t \bar{q}_\epsilon - \Delta \bar{q}_\epsilon = -(y_\epsilon + y_0) \quad \text{in } \Omega \times (0, T] \tag{8a}$$

$$\lim_{\epsilon \searrow 0} \frac{J(0, \hat{\chi}_{\omega_\epsilon} \bar{p}_\epsilon, f) - J(0, \hat{\chi}_\omega \bar{p}_\epsilon, f)}{|B_\epsilon(0)|} = \int_0^T \frac{1}{4\gamma} (\bar{p}(\eta_0, t))^2 \, dt. \tag{9}$$

$$J(0, u, f) = \tilde{G}(0, u, f, y^{u,f,\omega}, p^{u,f,\omega}). \tag{3.10}$$

$$J(\epsilon, \bar{u}_\epsilon, f) - J(0, \bar{u}_\epsilon, f) = -\mathrm{sgn}_\omega(\eta_0) \int_0^T \int_{B_\epsilon(\eta_0)} \bar{u}_\epsilon \bar{q}_\epsilon \, dx \, dt. \tag{A.11}$$

$$J(\epsilon, \bar{u}_\epsilon, f) - J(0, \bar{u}, f) = J(0, \bar{u}_\epsilon, f) - J(0, \bar{u}, f) - \mathrm{sgn}_\omega(\eta_0) \frac{1}{2\gamma} \int_0^T \int_{B_\epsilon(\eta_0)} \chi_{\omega_\epsilon} \bar{p}_\epsilon \bar{q}_\epsilon \, dx \, dt. \tag{12a}$$

$$\liminf_{\epsilon \searrow 0} \frac{J(\epsilon, \bar{u}_\epsilon, f) - J(0, \bar{u}, f)}{|B_\epsilon(\eta_0)|} \geq \frac{1}{4\gamma} \int_0^T (p^{\bar{u},f,\omega}(\eta_0, t))^2 \, dt. \tag{13}$$

$$\mathcal{G}(\epsilon, f) := \mathcal{J}(\omega_\epsilon, f), \quad \ell(\epsilon) := |B_\epsilon(\eta_0)|. \tag{7.14}$$

$$\lim_{t \searrow 0} \frac{\mathcal{G}(t, f) - \mathcal{G}(0, f)}{\ell(t)} =: \partial_\ell \mathcal{G}(0^+, \bar{f}) \tag{A.15}$$

$$\lim_{k \to \infty} \frac{\mathcal{G}(t_{n_k}, \bar{f}_{t_{n_k}}) - \mathcal{G}(0, \bar{f}_{t_{n_k}})}{\ell(t_{n_k})} = \partial_\ell \mathcal{G}(0^+, \bar{f}). \tag{16a}$$

$$\frac{d}{d\ell} g_1|_{t=0} = \lim_{\epsilon \searrow 0} \frac{g_1(\epsilon) - g_1(0)}{|B_\epsilon(0)|} = \max_{f \in \mathfrak{V}_1(0)} \mathcal{T} \mathcal{J}_1(\omega, f). \tag{17}$$

$$G(\epsilon, \bar{u}^{f,\omega_\epsilon}, f) - G(0, \bar{u}^{f,\omega_\epsilon}, f) = -\mathrm{sgn}_\omega(\eta_0) \int_0^T \int_{B_\epsilon(\eta_0)} \bar{u}^{f,\omega_\epsilon} \bar{q}^{\bar{u},f,\omega_\epsilon} \, dx \, dt, \tag{4.18}$$

$$\bar{u}^{f,\omega_\epsilon}(t) := \bar{u}_\epsilon(t) = \frac{1}{2|\Omega|\gamma} \int_{\omega_\epsilon} \bar{p}^{\bar{u}_\epsilon, f, \omega_\epsilon}(t,x)\, dx \tag{A.19}$$

$$\lim_{\epsilon \searrow 0} \frac{G(\epsilon, \bar{u}^{\omega,f}, f) - G(0, \bar{u}^{\omega,f}, f)}{|B_\epsilon(\eta_0)|} = -\mathrm{sgn}_\omega(\eta_0) \int_0^T \bar{u}^{\omega,f}(t) \bar{p}^{\bar{u},f,\omega}(t,\eta_0)\, dt. \tag{20a}$$

$$\partial_\ell G(0^+, \bar{u}^{\omega,f}, f) = -\mathrm{sgn}_\omega(\eta_0) \int_0^T \bar{u}^{f,\omega}(t) \bar{p}^{\bar{u},f,\omega}(t,\eta_0)\, dt. \tag{21}$$

$$\lim_{k\to\infty} \frac{G(\epsilon_{n_k}, \bar{u}_{\epsilon_{n_k}}, f_{n_k}) - G(0, \bar{u}_{\epsilon_{n_k}}, f_{n_k})}{|B_{\epsilon_{n_k}}(\eta_0)|} = -\mathrm{sgn}_\omega(\eta_0) \int_0^T \bar{u}^{\omega,f}(t) \bar{p}^{\bar{u},f,\omega}(t,\eta_0)\, dt. \tag{1.22}$$

$$f_h = \sum_{j=1}^N f_j \phi_j\,, \quad f_j \in \mathbf{R}\,, \phi_j \in V_h\,, \tag{A.23}$$

$$\begin{aligned} \partial_t y - \Delta y &= \chi_\omega u && \text{in } \Omega \times (0,T], \\ y &= 0 && \text{on } \partial\Omega \times (0,T], \\ y(0) &= f && \text{on } \Omega. \end{aligned} \tag{24a}$$

$$\varphi \circ T_t \to \varphi \quad \text{ strongly in } L_2(\Omega). \tag{25}$$

$$\frac{\varphi_n \circ T_{t_n} - \varphi_n}{t_n} \rightharpoonup \nabla\varphi \cdot X \quad \text{ weakly in } L_2(\Omega). \tag{5.26}$$

$$\mathcal{J}_1(\omega, f) = \inf_{u\in \mathrm{U}} J(\omega, u, f) = \int_0^T \|y(t)\|^2_{L^2(\Omega)} + \gamma |u(t)|^2\, dt + \alpha(|\omega| - c)^2, \quad \alpha > 0, \tag{A.27}$$

$$A(\tau) := \det(\partial T_\tau) \partial T_\tau^{-1} \partial T_\tau^{-\top}, \qquad \xi(\tau) := |\det(\partial T_\tau)|. \tag{28a}$$

$$-\frac{d}{dt}\Pi_h = A_h \Pi_h + \Pi_h A_h - \Pi_h B_h \gamma^{-1} B_h^\top \Pi_h + M_h \quad \text{ in } [0,T), \quad \Pi_h(T) = 0\,. \tag{29}$$

$$A(\tau,\cdot), \partial_\tau A(\tau,\cdot) \in L_\infty(\Omega, \mathbf{R}^{d\times d}), \quad \xi(\tau,\cdot), \partial_\tau \xi(\tau,\cdot) \in L_\infty(\Omega). \tag{2.30}$$

$$z_i[k] = \sum_{v_j \in \mathcal{N}_i^{\mathrm{mm}}[k]} \frac{1}{|\mathcal{S}_i^{\mathrm{mm}}[k]|} x_j[k] \tag{A.31}$$

$$[x_i[k+1]]_p = [z_i[k]]_p - \eta[k][g_i[k]]_p. \tag{32a}$$

$$\|z_i[k] - \hat{x}\|_2 \le \sum_{v_j \in \mathcal{N}_i^{\mathrm{mm}}[k]} \frac{1}{|\mathcal{S}_i^{\mathrm{mm}}[k]|} \|x_j[k] - \hat{x}\|_2 \le R_i[k]. \tag{33}$$

$$\min_{x\in\mathbb{R}^d} \frac{1}{N} \sum_{v_i \in \mathcal{V}} f_i(x) \tag{6.34}$$

$$D_{ij}[k] \triangleq \|x_j[k] - \hat{x}\|_2 \;\text{ for }\; x_j[k] \in \mathcal{S}_i[k] \setminus \{x_i[k]\}. \tag{A.35}$$

$$\mathcal{C}_i(\epsilon) \triangleq \{x \in \mathbb{R}^d : f_i(x) \leq f_i(x_i^*) + \epsilon\}. \tag{36a}$$

$$-\langle g(x),\ x^* - x\rangle \geq f(x) - f(x^*). \tag{37}$$

$$\cos\hat{\theta}(x) \geq \frac{\epsilon}{L\delta(\epsilon)}. \tag{3.38}$$

$$\max_{v_i\in\mathcal{R}} \|z_i[k] - \hat{x}\|_2 \leq \max_{v_i\in\mathcal{R}} \|x_i[k] - \hat{x}\|_2. \tag{A.39}$$

$$\Gamma_i(p, l) \triangleq p^2 - \Delta_i(p, l). \tag{40a}$$

$$\Gamma_i(s^*(\xi),\ \eta[k]\ \|g_i[k]\|_2) \leq \big(s^*(\xi)\big)^2. \tag{41}$$

$$\Delta_i(\|z_i[k] - \hat{x}\|_2, \eta[k]\ \|g_i[k]\|_2) > \Delta_i(s^*(\xi), \eta[k]\ \|g_i[k]\|_2). \tag{7.42}$$

$$\eta[k]\kappa_i \leq \eta[k]\ \|g_i[k]\|_2 \leq \eta[k]L \leq \frac{b_i}{2}. \tag{A.43}$$

$$\Delta_i\big(s^*(\xi),\ \eta[k]\ \|g_i[k]\|_2\big) \geq \frac{b_i}{2}\eta[k]\ \|g_i[k]\|_2. \tag{44a}$$

$$\max_{v_i\in\mathcal{R}} \|z_i[k^*] - \hat{x}\|_2 \leq \max_{v_i\in\mathcal{R}} \|x_i[k^*] - \hat{x}\|_2 \leq s^*(\xi). \tag{45}$$

$$\|g(x)\|_2\ \|x^* - x\|_2 \cos\hat{\theta}(x) \geq f(x) - f(x^*). \tag{4.46}$$

$$\hat{\theta}(x) \leq \arccos\Big(\frac{\epsilon}{L\delta(\epsilon)}\Big) \triangleq \theta(\epsilon) < \frac{\pi}{2}. \tag{A.47}$$

$$\begin{aligned}\Big\langle \sum_{v_i\in\mathcal{R}} g_i(x_0),\ u\Big\rangle &= \sum_{v_i\in\mathcal{R}} \langle g_i(x_0),\ u\rangle \\ &= \sum_{v_i\in\mathcal{R}} \|g_i(x_0)\|_2 \cos\angle(g_i(x_0),\ x_0 - \hat{x}) \\ &> 0\end{aligned} \tag{48a}$$

$$\max_{y\in\mathcal{B}(\hat{x},R)} \angle(x - y,\ x - \hat{x}) = \arcsin\Big(\frac{R}{\|x - \hat{x}\|_2}\Big). \tag{49}$$

$$\begin{aligned}\|x_i[k+1] - \hat{x}\|_2 &= \|z_i[k] - \hat{x} - \eta[k]g_i[k]\|_2 \\ &\leq \|z_i[k] - \hat{x}\|_2 + \eta[k]\|g_i[k]\|_2.\end{aligned} \tag{1.50}$$

$$\eta[k_2^*] \leq \frac{1}{L}\min_{v_i\in\mathcal{R}} \big\{\min\{a_i^+,\ b_i\}\big\}. \tag{A.51}$$

$$\eta[k]\ \|g_i[k]\|_2 \leq \eta[k]L \leq \min_{v_j\in\mathcal{R}} \big\{\min\{a_j^+,\ b_j\}\big\} \leq a_i^+. \tag{52a}$$

$$\mathcal{I}_x[k] \triangleq \{v_i \in \mathcal{R} : \|x_i[k] - \hat{x}\|_2 > s^*(\xi)\} \tag{53}$$

$$\mathcal{I}_z[k] \triangleq \{v_i \in \mathcal{R} : \|z_i[k] - \hat{x}\|_2 > s^*(\xi)\}. \tag{5.54}$$

$$\begin{aligned}\max_{v_i\in\mathcal{I}_z[k]}\|x_i[k+1]-\hat{x}\|_2^2 &\geq \max_{v_i\in\mathcal{I}_x[k+1]}\|x_i[k+1]-\hat{x}\|_2^2\\ &= \max_{v_i\in\mathcal{R}}\|x_i[k+1]-\hat{x}\|_2^2.\end{aligned} \tag{A.55}$$

$$\begin{aligned}\|z_i[k]-\hat{x}\|_2 &\leq \max_{v_j\in(\mathcal{N}_i^-\cap\mathcal{R})\cup\{v_i\}}\|x_j[k]-\hat{x}\|_2\\ &\leq \max_{v_j\in\mathcal{R}}\|x_j[k]-\hat{x}\|_2,\end{aligned} \tag{56a}$$

$$\max_{v_i\in\mathcal{R}}\|x_i[k^*+K^*]-\hat{x}\|_2^2 \leq \big(s^*(\xi)\big)^2. \tag{57}$$

$$\|x_i[k^*]-\hat{x}\|_2 \leq s^*(\xi) \tag{2.58}$$

$$\max_{v_i\in\mathcal{R}}\|x_i[k^*]-\hat{x}\|_2 \leq s^*(\xi). \tag{A.59}$$

$$\max_{v_i\in\mathcal{R}}\|x_i[k^*+K]-\hat{x}\|_2 \leq s^*(\xi) \quad \text{for all } \ K\geq 0 \tag{60a}$$

$$\rho=\sum_{i=1}^{r}\rho_i^{(1)}\otimes\cdots\otimes\rho_i^{(n)} \quad \text{with all } \rho_i^{(j)}\succcurlyeq 0. \tag{61}$$

$$\operatorname{tr}((\sigma\otimes\tau)C_T)\geqslant 0 \quad \text{for all } \sigma\succcurlyeq 0,\tau\succcurlyeq 0. \tag{6.62}$$

$$\tau_i\in C_{t_i} \quad \text{with} \quad i=1,\ldots,n, \tag{A.63}$$

$$v^*\tau v=\sum_{i,j}\bar{\lambda}_i\lambda_j\mu_{i,j}. \tag{64a}$$

$$p=\sum_{\omega}\sigma_\omega\otimes\omega, \tag{65}$$

$$p=\sum_{i=1}^{d}E_{ii}\otimes z_i \tag{3.66}$$

$$(\tau_1\oplus\chi_1,\ldots,\tau_d\oplus\chi_d)\in C_{r+s}(p), \tag{A.67}$$

$$\ell=\sigma_0\otimes 1+\sum_{i=1}^{d}\sigma_i\otimes z_i \tag{68a}$$

$$(\tau_1,\tau_2)\in C_s(\ell). \tag{69}$$

$$0\preccurlyeq\sigma_i\preccurlyeq I_m, \tag{7.70}$$

$$\ell:=I_m\otimes 1-\sum_{i=1}^{d}(2\sigma_i-I_m)\otimes z_i \tag{A.71}$$

$$v^*\sigma_i v=\tau_i \text{ for all } i=1,\ldots,d. \tag{72a}$$

$$w\colon \mathcal{Q}_A \times \mathcal{Q}_B \times \mathcal{A}_A \times \mathcal{A}_B \to \{0,1\} \tag{73}$$

$$c_A\colon \mathcal{Q}_A \to \mathcal{A}_A \quad \text{and} \quad c_B\colon \mathcal{Q}_B \to \mathcal{A}_B, \tag{4.74}$$

$$p(x,y \mid a,b) = \sum_{\lambda} q_\lambda \, r_A(a,\lambda)(x) \cdot r_B(b,\lambda)(y), \tag{A.75}$$

$$\mathcal{G} := \mathbb{C}\langle \mathcal{Q}_A \times \mathcal{A}_A, \mathcal{Q}_B \times \mathcal{A}_B \rangle. \tag{76a}$$

$$\varphi\colon \mathcal{G} \to \mathbb{C} \tag{77}$$

$$\rho = \sum_{i_1,\ldots,i_n=1}^{r} \rho^{(1)}_{i_1,i_2} \otimes \rho^{(2)}_{i_2,i_3} \otimes \cdots \otimes \rho^{(n)}_{i_n,i_1}. \tag{1.78}$$

$$\rho = \xi\xi^* \quad \text{with} \quad \xi = \sum_{i_1,\ldots,i_n=1}^{r} \xi^{(1)}_{i_1,i_2} \otimes \xi^{(2)}_{i_2,i_3} \otimes \cdots \otimes \xi^{(n)}_{i_n,i_1}. \tag{A.79}$$

$$\rho = \sum_{i_1,\ldots,i_n=1}^{r} \rho_{i_1,i_2} \otimes \rho_{i_2,i_3} \otimes \cdots \otimes \rho_{i_n,i_1}, \tag{80a}$$

$$\rho = \sum_{\alpha} \rho_\alpha \otimes \rho_\alpha \quad \text{with all } \rho_\alpha \text{ psd}, \tag{81}$$

$$\rho = \sum_{\alpha,\beta} \rho_{\alpha,\beta} \otimes \rho_{\beta,\alpha} \quad \text{with all } \rho_{\alpha,\beta} \text{ psd}. \tag{5.82}$$

$$\rho = \sum_{i} \rho^{(1)}_i \otimes \cdots \otimes \rho^{(n)}_i \tag{A.83}$$

$$\mathbb{P}\left(|X - \mathbb{E}X| > \frac{1}{2}\mathbb{E}X\right) \leq \frac{4\mathrm{Var}(X)}{(\mathbb{E}X)^2} = 4\left(\frac{\mathbb{E}(X(X-1))}{(\mathbb{E}X)^2} + \frac{1}{\mathbb{E}X} - 1\right). \tag{84a}$$

$$\mathbb{E}\,|S(v,H_i;t)| \leq \frac{t^{w_i(v)}}{n^{w_i(v)-1}} \left(2(w_i(v)!)^D - 1\right), \tag{85}$$

$$\mathbb{E}\,|S(v,H_i;t)| \leq \frac{t^{w}}{n^{w-1}} \left(2w^{d_i(x,v)} \left((w-1)!\right)^D - 1\right). \tag{2.86}$$

$$\tau_{\mathcal{P}}(q,n) = \inf_{\mathcal{S}} \tau_{\mathcal{P}}(\mathcal{S},q,n), \tag{A.87}$$

$$\mathbb{E}\,|S(v,H_i;t)| \leq \frac{t^{w_i}}{n^{w_i-1}} \left(2\left((s-1)^{w_i} w_i!\right)^D - 1\right), \tag{88a}$$

$$\begin{aligned} \mathbb{E}\,|S(v,H_i;t)| &\leq \frac{t^{w}}{n^{w-1}} \left(2w^{d(x,u)+1}\left((w-1)!\right)^D - w + \frac{t}{n-t}\right) \\ &\leq \frac{t^{w}}{n^{w-1}} \left(2w^{d(x,v)}\left((w-1)!\right)^D - 1\right), \end{aligned} \tag{89}$$

$$\mathbb{E}\ |S(v,H_i;t)| \leq (s-1)w_i\mathbb{E}\ |S(u,H_i;t)| + 2\sum_{a\geq w_i+1}(s-1)a\frac{t^a}{a!n^{a-1}} \tag{6.90}$$

$$\lim_{n\to\infty}\mathbb{P}[G_t \in \mathcal{P}_H] = 1. \tag{A.91}$$

$$\lim_{n\to\infty}\mathbb{P}[G_t \in \mathcal{P}_H] = 0. \tag{92a}$$

$$\lim_{n\to\infty}\mathbb{P}[G_t^{(s)} \in \mathcal{P}_H] = 1. \tag{93}$$

$$\lim_{n\to\infty}\mathbb{P}[G_t^{(s)} \in \mathcal{P}_H] = 0. \tag{3.94}$$

$$X_t^{(x)} = \sum_{i=1}^{n} I_t(i)^{(x)}, \tag{A.95}$$

$$\mathbb{E}(X(X-1)) = n(n-1)\mathbb{P}(I_t(1)^{(x)} = I_t(2)^{(x)} = 1), \tag{96a}$$

$$\mathbb{P}(I_t(1)^{(x)} = I_t(2)^{(x)} = 1) = \binom{t}{x}\binom{t-x}{x}\left(\frac{1}{n}\right)^{2x}\left(1-\frac{2}{n}\right)^{t-2x} \sim \frac{t^{2x}}{x!^2n^{2x}}, \tag{97}$$

$$w_{H_i}(x_i) = w_{H_{i-1}}(x_i) + 1 \tag{7.98}$$

$$\mathbb{E}\ |S(v,H;t)| \leq \frac{t^w}{n^{w-1}}\left(2(m!)^D - 1\right) \tag{A.99}$$

$$S(v,H_0;t) = n = \frac{t^0}{n^{-1}}\left(2(0!)^D - 1\right), \tag{100a}$$

$$|S(v,H_i;t)| \leq |S(v,H_{i-1};t)|, \tag{101}$$

$$\mathbb{E}\ |S(x,H_i;t)| \leq \sum_{t'=0}^{t-1}\frac{|S(x,H_{i-1};t')|}{n} \tag{4.102}$$

$$\mathbb{E}\ |S(v,H';t)| \leq \frac{t^d}{n^{d-1}}\left(2(e(H')!)^{|V(H')|-1} - 1\right) = O\left(\frac{t^d}{n^{d-1}}\right). \tag{A.103}$$

$$\mathbb{E}\ |S(v,H;t)| \leq \frac{t^w}{n^{w-1}}\left(2\left((s-1)^m m!\right)^D - 1\right) \tag{104a}$$

$$\tau_{\mathcal{P}} := \lim_{q\to 1^-}\limsup_{n\to\infty}\frac{\ln(\tau_{\mathcal{P}}(q,n))}{\ln n} \tag{105}$$

$$[X_1] = [X_2] \iff [X_1 \cdot X_2^{-1}] = 0. \tag{1.106}$$

$$[C] = [0] \iff [\sigma_I(C)] = [0]. \tag{A.107}$$

$$[X_1 \cdot X_2 \cdot X_3 \cdot X_4] = [0] \iff [X_2 \cdot X_3 \cdot X_4 \cdot X_1] = [0] \tag{108a}$$

$$[X_1^{-1}] = [X_2 \cdot X_3 \cdot X_4] \iff [X_2^{-1}] = [X_3 \cdot X_4 \cdot X_1]. \tag{109}$$

$$[\phi(C)] = [\beta_{W_v}(\psi(C))]. \tag{5.110}$$

$$[\phi(U)] = [W_u \cdot \psi(U) \cdot W_v^{-1}]. \tag{A.111}$$

$$[C] = [C'] \text{ implies } [\![C]\!] = [\![C']\!]. \tag{112a}$$

$$(r)(r, x_1)(r, x_1, x_2) \dots (r, x_1, x_2 \dots, x_\ell). \tag{113}$$

$$\phi : \mathcal{U} \to H : (r, x_1, \dots, x_\ell) \mapsto x_\ell \tag{2.114}$$

$$\Phi : (r, x_1, \dots, x_\ell) \mapsto (\phi(r), \phi(r, x_1), \dots, \phi(r, x_1, \dots, x_\ell)) \tag{A.115}$$

$$\Phi^{-1}(X)_i := \Phi^{-1}(X)_{i-1} \cdot \phi^{-1}_{x_{i-1}}(x_i) \tag{116a}$$

$$W_d = \sqrt{A}^d \cdot T \cdot I \cdot S^{-1} \tag{117}$$

$$[\sqrt{A}^d \cdot T \cdot I \cdot S^{-1}] = [\sqrt{B}^{d'} \cdot T' \cdot I' \cdot S'^{-1}] \tag{6.118}$$

$$[T \cdot \sqrt{A_f}^d \cdot I \cdot S^{-1}] = [T' \cdot \sqrt{B}_f^{d'} \cdot I' \cdot S'^{-1}] \tag{A.119}$$

$$[\phi(C)] \neq [\beta_W(\psi(C))] \tag{120a}$$

$$W_d := \sqrt{\phi(C)}^d \cdot T \cdot I \cdot S^{-1} \tag{121}$$

$$[W_u] = [\phi(e) \cdot W_v \cdot \phi(e)^{-1}], \tag{3.122}$$

$$g(t) := \frac{1}{c + (k-1)t} \tag{A.123}$$

$$\begin{aligned} H =& \sum_{y=1}^{g} \prod_{i=1}^{k} (1 - \chi_{D'}(D'y + n' + h_i)\chi_g(D'y + n' + h_i)) \\ =& \sum_{y=1}^{g} \prod_{i=1}^{k} (1 - \chi_{D'}(n' + h_i)\chi_g(D'y + n' + h_i)) \end{aligned} \tag{124a}$$

$$H = \left| \left(\sum_{y=1}^{g} \prod_{i=1}^{k} \epsilon_i \right) + \left(\sum_{y=1}^{g} \sum_{j=1}^{2^k - 1} \left(\prod_{i \in S_j} \epsilon_i \right) \chi_g(Q_j(y)) \right) \right| \tag{125}$$

$$\begin{aligned} m_2 &:= \int_0^T g(t)^2 \, dt \\ \mu &:= \frac{1}{m_2} \int_0^T t g(t)^2 \, dt \\ \sigma^2 &:= \frac{1}{m_2} \int_0^T t^2 g(t)^2 \, dt - \mu^2. \end{aligned} \tag{7.126}$$

$$\pi(x,q,a) = \frac{\pi(x)}{\phi(q)}\left(1 - \chi_D\left(\frac{aD}{(q,D)}\right) + O\left(L(1,\chi_D)(\log x)^{r^r}\right)\right), \tag{A.127}$$

$$\begin{aligned} I_k(F) &= \int_0^\infty ... \int_0^\infty F(t_1,...,t_k)^2 dt_1...dt_k, \\ J_k^{(m)}(F) &= \int_0^\infty ... \left(\int_0^\infty F(t_1,...,t_k)^2 dt_m\right) dt_1...dt_{m-1}dt_{m+1}dt_k. \end{aligned} \tag{128a}$$

$$\left(\int_0^{r-X_1-\cdots-X_{k-1}} g(t)^2 dt\right) \leq \frac{m_2 M_k^{[T]} r}{k} \mathbb{P}\left(X_1 + \cdots + X_k \geq r\right) \tag{129}$$

$$\begin{aligned} H_1 &\leq 12, \\ H_2 &\leq 270, \\ H_3 &\leq 52,116, \\ H_4 &\leq 474,266, \\ H_5 &\leq 4,137,854, \\ H_m &\ll me^{2m} \end{aligned} \tag{4.130}$$

$$\begin{aligned} H_1 &= 2 \\ H_2 &\leq 12, \\ H_4 &\leq 270, \\ H_6 &\leq 52,116, \\ H_m &\ll e^{(1+\epsilon')m}, \end{aligned} \tag{A.131}$$

$$\begin{aligned} 5.9484 &> \frac{3}{.504346916} = 5.94828... \\ 7.9310494 &> \frac{4}{.504346916} = 7.9310487... \\ 9.9138119 &> \frac{5}{.504346916} = 9.9138109... \end{aligned} \tag{132a}$$

$$H_1 \leq 70,000,000, \tag{133}$$

$$L(s,\chi_q) = \sum_{n=1}^{\infty} \frac{\chi_q(n)}{n^s}. \tag{1.134}$$

$$\beta < 1 - C(\epsilon) q^{-\epsilon}. \tag{A.135}$$

$$s_D > 1 - \frac{1}{(\log D)^{r^r + A}}. \tag{136a}$$

$$q < x^{\frac{58\left(1-\frac{1}{r}\right)}{115}} = x^{.5043469...}, \tag{137}$$

$$M_k > \frac{2m}{\theta'}. \tag{5.138}$$

$$T = \{n + h_1, \cdots, n + h_k\}. \tag{A.139}$$

$$Re(s) > 1 - \frac{1}{(\log D)^{r^r+A}} > 1 - \frac{1}{(\log C)^{r^r+A}}. \tag{140a}$$

$$\prod_{i=1}^{k} \left(1 - \chi_D(x + n' + h_i)\right). \tag{141}$$

$$\left| \sum_{y=1}^{g} \chi_g(Q_j(y)) \right| \leq (k-1)\sqrt{g}. \tag{2.142}$$

$$l \equiv D'y' + n' \pmod{D} \tag{A.143}$$

$$\mathcal{R}_k = \{(t_1, ..., t_k) : \sum_{i=1}^{k} t_i \leq 1\} \tag{144a}$$

$$B = \frac{\phi(W)}{W} \log R, \tag{145}$$

$$S_2 = (\theta + o(1)) \frac{X}{W} B^{-k} \sum_{m=1}^{k} J_k^{(m)}(F). \tag{6.146}$$

$$S = \left((\theta + o(1)) \sum_{m=1}^{k} J_k^{(m)}(F) - \rho I_k(F) \right) \frac{X}{W} B^{-k}. \tag{A.147}$$

$$\pi(x, q, a) = \frac{2\pi(X)}{\phi(q)} + O\left(\frac{\pi(X)}{\phi(q) \log^{A-2}(x)} \right). \tag{148a}$$

$$(\theta - 2\epsilon) M_k > \rho. \tag{149}$$

$$M_k \geq \log k - 2 \log \log k - 2. \tag{3.150}$$

$$k \gg e^{(1.98276)\rho}. \tag{A.151}$$

$$\mathcal{R}_k^{[\alpha]} = \{(t_1, ..., t_k) \in [0, \alpha]^k : \sum_{i=1}^{k} t_i \leq 1\}. \tag{152a}$$

$$p \mapsto \langle V(p), W(p) \rangle_p \tag{153}$$

$$F(p) = 0_p \in T_p \mathcal{M}. \tag{7.154}$$

$$\operatorname{grad}_x \mathcal{L}(x, y, z) = 0_x; \ h(x) = 0, g(x) \leq 0, z \geq 0; \ Zg(x) = 0. \tag{A.155}$$

$$\bar{\mathrm{R}}_w(\xi) := (\mathrm{R}_x(\xi_x), y + \xi_y, z + \xi_z, s + \xi_s) \tag{156a}$$

$$\operatorname{Hess}_x \mathcal{L}(w) = \operatorname{Hess} f(x) + \sum_{i=1}^{l} y_i \operatorname{Hess} h_i(x) + \sum_{i=1}^{m} z_i \operatorname{Hess} g_i(x), \tag{157}$$

$$F_\mu(w) := F(w) - \mu\hat{e}, \text{ and } \hat{e} := \hat{e}(w) := (0_x, 0, 0, e)\,. \tag{4.158}$$

$$\nabla F_\mu(w) = \nabla F(w) - \mu\nabla\hat{e}(w) = \nabla F(w), \tag{A.159}$$

$$\nabla F(w_k)[\Delta w_k] = -F(w_k) + \mu_k\hat{e}; \tag{160a}$$

$$\mathcal{T}(\Delta x, \Delta y) := \begin{pmatrix} \mathcal{A}_w[\Delta x] + \mathcal{H}_x[\Delta y] \\ \mathcal{H}_x^*[\Delta x] \end{pmatrix} = \begin{pmatrix} c \\ q \end{pmatrix}. \tag{161}$$

$$\mathcal{A}_w[\Delta x] = c \tag{1.162}$$

$$\mathrm{T}_\eta\,(\xi) := \mathrm{P}_\gamma^{1\to 0}(\xi) \tag{A.163}$$

$$\nabla F(p_k)[\xi_k] = -F(p_k) + \mu_k\hat{e}; \tag{164a}$$

$$\|F(p_k)\| = O(d(p_k, p^*)). \tag{165}$$

$$\alpha^i := \max_{\alpha\in(0,1]} \left\{\alpha : f^i(t) \geq 0, \text{ for all } t \in (0, \alpha]\right\}, \tag{5.166}$$

$$\nabla F(w_k)[\Delta w_k] = -F(w_k) + \sigma_k\rho_k\hat{e}; \tag{A.167}$$

$$y = \left[\left(\mathcal{H}_x^*\mathcal{H}_x\right)^{-1}\mathcal{H}_x^*\right]\left(\mathrm{grad}_x\,\mathcal{L}(w) - \mathrm{grad}\,f(x) - \mathcal{G}_x z\right). \tag{168a}$$

$$\varphi'(\alpha) = \mathcal{D}\varphi(\bar{\mathrm{Exp}}_{w_k}(\alpha\Delta w_k))[\mathcal{D}\bar{\mathrm{Exp}}_{w_k}(\alpha\Delta w_k)\,[\Delta w_k]]. \tag{169}$$

$$\mathrm{grad}_x\,\mathcal{L}(x, y, z) = \mathrm{grad}\,f(x) + \sum_{i=1}^{l} y_i\,\mathrm{grad}\,h_i(x) + \sum_{i=1}^{m} z_i\,\mathrm{grad}\,g_i(x), \tag{2.170}$$

$$\left\|\mathrm{P}_\gamma^{0\to 1}[F(y)] - F(x) - \int_0^1 \mathrm{P}_\gamma^{0\to t}\nabla F(\gamma(t))\mathrm{P}_\gamma^{t\to 0}[\eta]\mathrm{d}t\right\| \leq c_1\|\eta\|^2, \tag{A.171}$$

$$p_{k+1} := \mathrm{R}_{p_k}[\alpha_k\nabla F(p_k)^{-1}(-F(p_k) + \mu_k\hat{e})], \tag{172a}$$

$$\Pi := 2\max\left\{\max_i\left\{1/(s^*)_i \mid (s^*)_i > 0\right\}, \max_i\left\{1/(z^*)_i \mid (z^*)_i > 0\right\}\right\}. \tag{173}$$

$$\tau_1 := \min(Z_0S_0e)/(z_0^Ts_0/m),\ \tau_2 := z_0^Ts_0/\left\|F(w_0)\right\|. \tag{6.174}$$

$$\langle\mathrm{grad}\,\varphi(w_k), \Delta w_k\rangle = 2(-\|F(w_k)\|^2 + \sigma_k\rho_k z_k^Ts_k). \tag{A.175}$$

$$\|\mathcal{A}_x y\|_x^2 = \|\sum_{i=1}^{n}(\hat{\mathcal{A}}_x\hat{y})_i E_i\|_x^2 = \sum_{i=1}^{n}(\hat{\mathcal{A}}_x\hat{y})_i^2 = \|\hat{\mathcal{A}}_x\hat{y}\|_2^2. \tag{176a}$$

$$\hat{\nabla}F(w_k) = \begin{pmatrix} Q_k & B_k & C_k & 0 \\ B_k^T & 0 & 0 & 0 \\ C_k^T & 0 & 0 & I \\ 0 & 0 & S_k & Z_k \end{pmatrix}. \tag{177}$$

$$\min_{X\in\mathbb{R}^{n\times k}} -2\,\mathrm{trace}(X^TC) \quad \text{s.t. } X^TX = I_k,\ X\geq 0. \tag{3.178}$$

$$\min_{X\in\mathbb{R}^{m\times n}} \|A-X\|_{\mathrm{F}}^2 \quad \text{s.t. } \mathrm{rank}(X) = r,\ X\geq 0. \tag{A.179}$$

$$d(p_{k+1},p^*) = (1-\alpha_k)O(d(p_k,p^*)) + O(d^2(p_k,p^*)) + O(\mu_k). \tag{180a}$$

$$S_k^{-1}\Delta s_k + Z_k^{-1}\Delta z_k = \mu_k(S_kZ_k)^{-1}e - e, \tag{181}$$

$$\|Q(w)\|_{\mathrm{F}} \leq \|\hat{\mathrm{Hess}} f(x)\|_{\mathrm{F}} + \sum_{i=1}^{l}|y_i|\|\hat{\mathrm{Hess}} h_i(x)\|_{\mathrm{F}} + \sum_{i=1}^{m}|z_i|\|\hat{\mathrm{Hess}} g_i(x)\|_{\mathrm{F}} =: T(w). \tag{7.182}$$

$$\lim_{t\to s+}\|u(t,\cdot;s,u_0) - u_0(\cdot)\|_{C^0(\bar{\Omega})} = 0, \tag{A.183}$$

$$\limsup_{t\to\infty}\int_\Omega u^{-p}(t,x;s,u_0)dx < \infty \tag{184a}$$

$$\lim_{t-s\to\infty}\int_\Omega u^{-p}(t,x;s,u_0) \leq M_1(p). \tag{185}$$

$$\|u(t,\cdot;s,u_0)\|_\infty \leq M_4^*(\theta) \quad \forall t-s\gg 1, \tag{4.186}$$

$$\int_\Omega \tilde{u}_0(x)dx \geq \frac{2}{3}\int_\Omega u_0(x)dx \quad \text{for any } \tilde{u}_0\in\mathcal{B}(u_0,r). \tag{A.187}$$

$$\int_\Omega u^{-1}(t,x;s,u_0)dx \leq \max\Big\{\int_\Omega u^{-1}(\tau,x;s,u_0)dx, \frac{b_{\sup}|\Omega|}{a_{\inf}-a_{\chi,\mu}}\Big\} \tag{188a}$$

$$\int_\Omega u^{-p}(t,x;s,u_0)dx \leq M_1^* \quad \forall\ s<t<\infty \tag{189}$$

$$\int_\Omega u^{q}(t,x;s,u_0)dx \leq M_2^* \quad \forall\, t\geq s. \tag{1.190}$$

$$\liminf_{t-s\to\infty}\inf_{x\in\Omega} u(t,x;s,u_n,a,b) \leq m_n := \frac{1}{n}. \tag{A.191}$$

$$u_1^*(x) = u(1,x;0,u_0^*,a^*,b^*). \tag{192a}$$

$$u(\cdot,t;0,u^*) = u^*(\cdot) \quad \text{for all } t\geq 0. \tag{193}$$

$$\int_\Omega \frac{|\nabla v(x,t;s,u_0)|^{2q}}{v^{k+1}(x,t;s,u_0)} \leq M(k,q)\int_\Omega \frac{u^q(x,t;s,u_0)}{v^{k-q+1}(x,t;s,u_0)} + M^*(k,q)\int_\Omega v^{2q-k-1}(x,t;s,u_0) \tag{5.194}$$

$$\int_\Omega u^q(t,x;s,u_0)dx \leq \max\Big\{\int_\Omega u^q(\tau,x;s,u_0)dx, M_2(p,q)+\tilde{M}_2(p,q,\tau,s,u_0)\Big\} \quad \forall\, t\geq\tau, \tag{A.195}$$

$$\|A_p^\beta e^{-A_pt}\|_{L^p(\Omega)} \leq C_{p,\beta}t^{-\beta}e^{-\gamma t} \ \text{ for } \ t>0. \tag{196a}$$

$$(\nu A^{-1}\mathcal{M}(\tilde{u}_0,a,b)u)(t,x) \geq \delta_0 \int_\Omega \mathcal{M}(\tilde{u}_0,a,b)u(x,t)dx \geq \frac{\delta_0}{2}\int_\Omega u_0(x)dx = \varepsilon \tag{197}$$

$$\frac{(p+1)\beta\mu}{p} - R < 0, \tag{2.198}$$

$$\frac{(p+1)\beta\mu}{p} = 2\beta\mu = 2(\chi+2-2\sqrt{\chi+1})\mu = a_{\chi,\mu} < a_{\inf}. \tag{A.199}$$

$$\frac{b_{\sup}|\Omega|(q-1)\Big\{(C^*(n,\nu))^{\frac{1}{q+1}}+\varepsilon\Big\}\max\{\chi,\chi^2\}}{4\delta_0(a_{\inf}-a_{\chi,\mu})} < b_{\inf}. \tag{200a}$$

$$|\Omega_\epsilon| = \int_\Omega u_n^{-p}\cdot u_n^p \leq \int_\Omega u_n^{-p}\cdot(2\epsilon)^p \leq (2\epsilon)^p\int_\Omega u_n^{-p} \leq (2\epsilon)^p M_1^*. \tag{201}$$

$$u(t+T,\cdot;0,u^T) = u(t,\cdot;T,u(T,\cdot;0,u^T)) = u(t,\cdot;0,u^T). \tag{6.202}$$

$$\mathcal{Y}(T) = \{a(\cdot,\cdot)\in\mathcal{X}_T \,:\, |a(t,x)-a(t^{'},x)| \leq B_1|t-t^{'}|^\gamma,\ \alpha\leq a(t,x)\leq B_2\ \forall\, t,t^{'}\in[s,s+T],\ x\in\bar{\Omega}\}, \tag{A.203}$$

$$\int_\Omega g(\lambda u_1(x)+(1-\lambda)u_2(x))dx \leq \lambda\int_\Omega g(u_1(x))dx + (1-\lambda)\int_\Omega g(u_2(x))dx. \tag{204a}$$

$$0 = \Delta v - \frac{1}{|\Omega|}\int_\Omega u(\cdot,t)+u, \quad x\in\Omega \tag{205}$$

$$(u(t,x;s,u_0,a,b),v(t,x;s,u_0,a,b)) = (u(t,x;s,u_0),v(t,x;s,u_0)). \tag{3.206}$$

$$\|u\|_{\mathcal{X}_T} = \max_{s\leq t\leq s+T}\|u(t,x)\|_{C^0(\bar{\Omega})}. \tag{A.207}$$

$$\lim_{t-s\to\infty}\int_\Omega u^{-1}(t,x;s,u_0)dx = \frac{b_{\sup}|\Omega|}{a_{\inf}-a_{\chi,\mu}}. \tag{208a}$$

$$2\beta - \frac{2N}{q} > \theta. \tag{209}$$

$$\lim_{n\to\infty} a_n(t,x) = a^*(t,x), \quad \lim_{n\to\infty} b_n(t,x) = b^*(t,x) \tag{7.210}$$

$$\liminf_{t-s\to\infty}\inf_{x\in\Omega} u(t,x;s,u_0,a,b) \geq m^*. \tag{A.211}$$

$$0\leq u_n(x)\leq 2\varepsilon \quad \forall\, x\in\Omega_\varepsilon. \tag{212a}$$

$$\mathcal{L}_{QED} = \bar{\psi}(i\ \not{\partial} - m)\psi - \bar{\psi}\psi\gamma_5\psi - e\bar{\psi}A\psi \tag{213}$$

$$S_{eff}[b,A] = -i\,\mathrm{Tr}\ln(\not{p} - m - \not{\beta}\gamma_5 - e\ \not{A}) \tag{4.214}$$

$$S'_{eff}[b,A] = i\,\mathrm{Tr}\sum_{n=1}^{\infty}\frac{1}{n}\left[\frac{1}{\not{p}-m-\not{p}\gamma_5}e\ \not{A}\right]^n. \tag{A.215}$$

$$S'_{eff}[b,A] = S^{(2)}_{eff}[b,A] + \dots \tag{216a}$$

$$S^{(2)}_{\text{eff}}[b,A] = \frac{ie^2}{2}\,\text{Tr}\left[\frac{1}{\not p - m - \not\beta\gamma_5} A\infty \frac{1}{\not - m - \not p\gamma_5} A\right]. \tag{217}$$

$$S^{(2)}_{\text{eff}}[b,A(x)] = \frac{1}{2}\int d^4x \Pi^{\alpha\mu\nu} F_{\alpha\mu} A_\nu \tag{1.218}$$

$$\Pi^{\alpha\mu\nu} = -\frac{ie^2}{2}\int \frac{d^4p}{(2\pi)^4}\,\text{tr}\left[S_b(p)\gamma^\mu S_b(p)\gamma^\alpha S_b(p)\gamma^\nu\right], \tag{A.219}$$

$$S_b(p) = \frac{i}{\not p - m - \not p\gamma_5} \tag{220a}$$

$$S_b(p) = i\left[\frac{\not p + m - \gamma_5\ \not b}{(p^2-m^2)} - \frac{2\gamma_5(m\ \not p - (b\cdot p))(\not p + m)}{(p^2-m^2)^2}\right] + \cdots. \tag{221}$$

$$p_\rho \to \vec{p}_\rho + p_0\delta_{0\rho} \tag{5.222}$$

$$S^{\text{cov}}_{\text{eff}} = \frac{(-i)}{2}\int d^4x I_1 \varepsilon^{\alpha\mu\nu\beta} b_\beta F_{\alpha\mu} A_\nu \tag{A.223}$$

$$I_1 = \frac{-3ie^2}{2\pi}\int \frac{d^D\vec{p}}{(2\pi)^D}\int_{-\infty}^{+\infty} dp_0 \left[\frac{\left(1-\frac{4}{D}\right)\vec{p}^2 + p_0^2 - m^2}{\left(\vec{p}^2 + p_0^2 + m^2\right)^3}\right] \tag{224a}$$

$$I_2 = \frac{ie^2}{\pi}\int \frac{d^D\vec{p}}{(2\pi)^D}\int_{-\infty}^{+\infty} dp_0 \left[\frac{\vec{p}^2 - p_0^2}{\left(\vec{p}^2 + p_0^2 + m^2\right)^3}\right]. \tag{225}$$

$$I_1 = \frac{3ie^2}{8D}\int \frac{d^D\vec{p}}{(2\pi)^D}\left[\frac{2(3-D)\vec{p}^2 + Dm^2}{\left(\vec{p}^2 + m^2\right)^{5/2}}\right] \tag{2.226}$$

$$I_2 = -\frac{ie^2}{8D}\int \frac{d^D\vec{p}}{(2\pi)^D}\left[\frac{(3-D)\vec{p}^2 - Dm^2}{\left(\vec{p}^2 + m^2\right)^{5/2}}\right] \tag{A.227}$$

$$I_1 = \frac{3ie^2}{8(4\pi)^{D/2}}\frac{\Gamma\left(1+\frac{\epsilon}{2}\right)}{\Gamma\left(\frac{5}{2}\right)\left(m^2\right)^{\epsilon/2}} \tag{228a}$$

$$S^{\text{cov}}_{\text{eff}} = \frac{1}{2}\int d^4x \varepsilon^{\alpha\mu\nu\beta} k_\beta F_{\alpha\mu} A_\nu \tag{229}$$

$$k_\beta = \frac{e^2}{16\pi^2} b_\beta \tag{6.230}$$

$$S_b(p) = \frac{i}{\not p - m} + \frac{i}{\not p - m}\left(-ip\gamma\gamma_5\right)\frac{i}{\not p - m} + \cdots \tag{A.231}$$

$$\Pi_{\mathbf{e}}^{\mu\alpha\nu} = -2ie^2\epsilon^{\mu\alpha\nu\rho}\int\frac{d^4p}{(2\pi)^4}\frac{1}{\left(p^2+m^2\right)^3} \times\left[b_\rho\left(p^2-3m^2\right)-4p_\rho(b\cdot p)\right], \tag{232a}$$

$$\tilde{k}_\beta = \frac{e^2}{4\pi^2}b_\beta. \tag{233}$$

$$\tilde{k}_\beta = k_\beta + \frac{3e^2}{16\pi^2}b_\beta \tag{3.234}$$

$$\Delta k_\beta = \frac{3e^2}{16\pi^2}b_\beta \tag{A.235}$$

$$S'_{eff}[b,A] = i\,\mathrm{Tr}\sum_{n=1}^{\infty}\frac{1}{n}\left[\frac{1}{\not p - m - e\ \not A}\ \not\gamma_5\right]^n. \tag{236a}$$

$$S^{(1)}_{eff}[b,A] = i\,\mathrm{Tr}\left[\frac{1}{\not p - m - eA}\ \not\gamma_5\right]. \tag{237}$$

$$\frac{1}{A-B} = \frac{1}{A} + \frac{1}{A}B\frac{1}{A} + \frac{1}{A}B\frac{1}{A}B\frac{1}{A} + \cdots \tag{7.238}$$

$$S^{(1)}_{eff}[b,A] = ie^2\,\mathrm{Tr}\left[\frac{1}{p-m}\ \not\gamma_5\frac{1}{p-m}A\frac{1}{p-m}A\right], \tag{A.239}$$

$$S^{(1)}_{\mathrm{eff}}[b,A(x)] = \frac{1}{2}\int d^4x\Pi^{\alpha\mu\nu}F_{\alpha\mu}A_\nu, \tag{240a}$$

$$\int\frac{d^4p}{(2\pi)^4}p_\mu p_\nu f\left(p^2\right) = \frac{g_{\mu\nu}}{4}\int\frac{d^4p}{(2\pi)^4}p^2 f\left(p^2\right), \tag{241}$$

$$\Pi^{\mu\alpha\nu} = \frac{e^2}{16\pi^2}\varepsilon^{\mu\alpha\nu\theta}b_\theta \tag{4.242}$$

$$\rho^2 = \rho\ . \tag{A.243}$$

$$\mathrm{Tr}\,\rho\ =\ \sum_k \mathbb{P}(k)\ =\ 1\ , \tag{244a}$$

$$\frac{d\rho}{dt}\ =\ -i\left(\mathrm{H}_{\mathrm{eff}}\,\rho - \rho\,\mathrm{H}^\dagger_{\mathrm{eff}}\right)\ =\ -i\left[\mathrm{E},\rho\right] - \frac{1}{2}\left\{\Gamma,\rho\right\}. \tag{245}$$

$$\frac{d}{dt}\mathrm{Tr}\,\rho\ =\ \frac{d}{dt}\sum_k\mathbb{P}(k)\ =\ -\mathrm{Tr}\big(\Gamma\rho\big)\ \neq\ 0\ . \tag{1.246}$$

$$\mathbb{P}(k|\mathcal{H}_1)\ =\ \frac{\mathbb{P}(k)}{\mathbb{P}(\mathcal{H}_1)}\ =\ \frac{\mathbb{P}(k)}{\sum_{q\in\mathcal{H}_1}\mathbb{P}(q)}\ =\ \frac{\mathbb{P}(k)}{\mathrm{Tr}\,\rho}\ , \tag{A.247}$$

$$\rho = \frac{1}{2}a_\mu\overline{\sigma}^\mu, \qquad E = E_\mu\sigma^\mu, \qquad \Gamma = \Gamma_\mu\sigma^\mu, \tag{248a}$$

$$\frac{da^0}{dt} = -\Gamma_\mu a^\mu, \tag{249}$$

$$\tau \,=\, |\mathbf{\Gamma}|\, t, \qquad r \,=\, \frac{|\mathbf{\Gamma}|}{2\,|\mathbf{E}|}\,, \tag{5.250}$$

$$\frac{d\mathbf{b}}{d\tau} \,=\, -\frac{1}{r}(\mathbf{e}\times\mathbf{b}) \,+\, \boldsymbol{\gamma} \,-\, (\boldsymbol{\gamma}\cdot\mathbf{b})\,\mathbf{b}\,. \tag{A.251}$$

$$\cos\theta_{\mathbf{b}_\star\boldsymbol{\gamma}} \,=\, \boldsymbol{\gamma}\cdot\mathbf{b}_\star \,=\, \frac{c_\gamma}{\alpha}\,. \tag{252a}$$

$$\lim_{r\to 0}\theta_{\mathbf{b}_\star\boldsymbol{\gamma}} = \theta_{\mathbf{e}\boldsymbol{\gamma}}, \qquad \lim_{r\to\infty}\theta_{\mathbf{b}_\star\boldsymbol{\gamma}} = 0\,. \tag{253}$$

$$X \,=\, \begin{pmatrix} p & p \\ -q & q \end{pmatrix}\,, \tag{2.254}$$

$$\frac{d\rho}{dt} \,=\, -i\,[\mathrm{E},\rho] \,-\, \frac{1}{2}\,\{\Gamma,\rho\} \,-\, [\mathrm{D},[\mathrm{D},\rho]]\,, \tag{A.255}$$

$$\frac{d|\mathbf{b}|^2}{d\tau} \,=\, 2\big(1-|\mathbf{b}|^2\big)\,|\mathbf{b}|\cos\theta_{\mathbf{b}\boldsymbol{\gamma}} \,-\, 2\zeta|\mathbf{b}|^2\sin^2\theta_{\mathbf{b}\mathbf{d}}\,. \tag{256a}$$

$$\mathbf{b}_\star \;\to\; \mathbf{b}_\star + \zeta\mathbb{J}^{-1}\mathbf{b}_\star - \zeta(\mathbf{d}\cdot\mathbf{b}_\star)\mathbb{J}^{-1}\mathbf{d}\,. \tag{257}$$

$$S[\rho] \,=\, -\,a^0\ln a^0 \,+\, a^0\,S[\hat\rho] \,=\, -\,a^0\ln a^0 \,-\, a^0\,\mathrm{Tr}\big(\hat\rho\ln\hat\rho\big)\,, \tag{6.258}$$

$$S[\hat\rho] \,=\, \ln 2 - \frac{1}{2}\,(1-|\mathbf{b}|)\ln(1-|\mathbf{b}|) - \frac{1}{2}\,(1+|\mathbf{b}|)\ln(1+|\mathbf{b}|)\,. \tag{A.259}$$

$$S[\hat\rho_{\mathrm{Coh}}] \,=\, 0\,, \qquad S[\hat\rho_{\mathrm{Mix}}] \,=\, \ln 2\,, \tag{260a}$$

$$\mathcal{T} \;\equiv\; \frac{(\mathbf{b}'\times\mathbf{b}'')\cdot\mathbf{b}'''}{|\mathbf{b}'\times\mathbf{b}''|^2} \;=\; 0, \tag{261}$$

$$\boldsymbol{\gamma} = \cos\varphi\;\hat{\mathbf{R}} - \sin\varphi\;\hat{\boldsymbol{\varphi}}, \qquad \mathbf{e}\times\boldsymbol{\gamma} = \sin\varphi\;\hat{\mathbf{R}} + \cos\varphi\;\hat{\boldsymbol{\varphi}}, \qquad \mathbf{e} = \hat{\mathbf{z}}. \tag{3.262}$$

$$\frac{d\varphi}{d\tau} \;=\; -\,\frac{1}{r} \,-\, \frac{\sin\varphi}{|\mathbf{b}|}\,. \tag{A.263}$$

$$|\mathbf{b}(\tau)| \;=\; \sqrt{1 \,-\, \mathcal{A}\,e^{2r\sin\varphi(\tau)}}\,, \tag{264a}$$

$$\widehat{\mathrm{P}} \;\simeq\; \frac{2\pi r}{1-\mathcal{A}}\left\{\left[1-\left(\frac{2-2\mathcal{A}}{2-\mathcal{A}}\right)^2 r^2\right]^{-1/2} - \,\mathcal{A}\right\}. \tag{265}$$

$$|\mathbf{b}| \,=\, 1\,, \qquad \widehat{\mathrm{P}} \,=\, \frac{2\pi r}{\sqrt{1-r^2}}\,. \tag{7.266}$$

$$\mathbf{b}_\star = \left(\sqrt{1-\frac{1}{r^2}+\frac{\zeta^2}{4}} - \frac{\zeta}{2}\right)\boldsymbol{\gamma} - \frac{1}{r}\mathbf{e}\times\boldsymbol{\gamma}. \tag{A.267}$$

$$\mathbf{b}_\star \;=\; -\,r\,\mathbf{e}\times\boldsymbol{\gamma}\,. \tag{268a}$$

$$\mathbf{n}_1=\mathbf{e},\qquad \mathbf{n}_2=\frac{1}{s_\gamma}\mathbf{e}\times\boldsymbol{\gamma},\qquad \mathbf{n}_3=\frac{1}{s_\gamma}\mathbf{e}\times(\mathbf{e}\times\boldsymbol{\gamma}), \tag{269}$$

$$\mathbf{b}=\alpha\mathbf{n}_1+\beta\mathbf{n}_2+\eta\mathbf{n}_3, \tag{4.270}$$

$$\alpha^4-(1-r^2)\alpha^2-c_\gamma^2 r^2=0\,. \tag{A.271}$$

$$\alpha \;=\; \pm\frac{1}{\sqrt{2}}\sqrt{1-r^2+\sqrt{(1-r^2)^2+4c_\gamma^2 r^2}}\,. \tag{272a}$$

$$S_{\rm AHM}[\phi,A_\mu]=\int \mathrm{d}^3x\left[-\frac{1}{4}F_{\mu\nu}^2+|D_\mu\phi|^2-\frac{\mu^2}{8}\left(|\phi|^2-v^2\right)^2\right], \tag{273}$$

$$\phi_\infty : S^1_\infty \to U(1)\cong S^1, \tag{1.274}$$

$$m_\sigma = m_A \equiv m. \tag{A.275}$$

$$\rho(r)=\left(v^2-\frac{1}{e}\frac{A'(r)}{r}\right)^{1/2}, \tag{276a}$$

$$\left(\frac{d^2}{d\xi^2}-\frac{1}{\xi}\frac{d}{d\xi}-1\right)G(\xi,\xi')=\delta\left(\xi-\xi'\right), \tag{277}$$

$$Z_N=N+\frac{\pi}{3\sqrt{3}}N^2+\frac{\pi^2}{108}N^3+\mathcal{O}\left(N^4\right). \tag{5.278}$$

$$R_{\rm core}\equiv\sqrt{2\times\frac{\int \mathrm{d}^2\mathbf{x}|\mathbf{x}|^2\mathcal{H}(\mathbf{x})}{\int \mathrm{d}^2\mathbf{x}\mathcal{H}(\mathbf{x})}}, \tag{A.279}$$

$$R_{\rm core}\ll R_{\rm interaction}, \tag{280a}$$

$$S_{\rm REFT}\left[\Phi,\sigma,A_\mu\right]=S_{\rm AHM}\left[\sigma,A_\mu\right]+S_{\rm vortex}\left[\Phi,\sigma,A_\mu\right]. \tag{281}$$

$$\overline{\Sigma}=\tilde{\Psi}^+ +\eta\overline{\Phi}. \tag{2.282}$$

$$c_{\Gamma\Gamma\Gamma}=\sqrt{\pi}M^{-1/2}N^{1/2}. \tag{A.283}$$

$$M\sim\mathbf{p}\sim\mathfrak{m}\gg\mathbf{q}\sim m. \tag{284a}$$

$$\int\frac{\mathrm{d}^{d-1}\mathbf{l}}{(2\pi)^{d-1}}\frac{\mathrm{d}\omega}{2\pi}\frac{\mathcal{N}(\omega,\mathbf{l})}{[\omega^2-\mathbf{l}^2-m^2+i0][\omega^2-(\mathbf{l}+\mathbf{q})^2-m^2+i0]}, \tag{285}$$

$$H_{\rm NREFT}\left(\mathbf{p}_1,\mathbf{p}_2,\mathbf{x}_1,\mathbf{x}_2\right)=\sum_{i=1}^{2}\sqrt{\mathbf{p}_i^2+M^2}+V\left(\mathbf{p}_1,\mathbf{p}_2,\mathbf{x}_1,\mathbf{x}_2\right). \tag{6.286}$$

$$V\left(\mathbf{p},\mathbf{p}'\right)=\sum_{i=1}^{\infty}V^{(i)}\left(\mathbf{p},\mathbf{p}'\right), \tag{A.287}$$

$$(\omega,\mathbf{l})\sim\left(|\mathbf{p}||\mathbf{q}|,|\mathbf{p}|\right). \tag{288a}$$

$$\mathcal{M}^{(1)}=-\frac{32\pi MN\left(M^2-(p_1\cdot p_2)\right)}{q^2-m^2}+\frac{8\pi MNm^2}{q^2-m^2}+\frac{32\pi MN}{m^2}\left(p_1\cdot p_2+\frac{1}{4}\left(q^2+m^2\right)\right). \tag{289}$$

$$V^{(1)}\left(\mathbf{p},\mathbf{x}\right)=\frac{8MN\mathbf{p}^2}{\mathbf{p}^2+M^2}K_0\left(mr\right). \tag{3.290}$$

$$g_{\sigma\sigma}=-2\pi. \tag{A.291}$$

$$L(\dot{\mathbf{x}}_1,\mathbf{x}_1;\dot{\mathbf{x}}_2,\mathbf{x}_2)=\frac{1}{2}M\dot{\mathbf{x}}_1^2+\frac{1}{2}M\dot{\mathbf{x}}_2^2-\tilde{U}\left(r_{12}\right)|\dot{\mathbf{x}}_1-\dot{\mathbf{x}}_2|^2, \tag{292a}$$

$$\eta_{\mu\nu}=\eta^{\mu\nu}=\begin{pmatrix}1&0&0\\0&-1&0\\0&0&-1\end{pmatrix}. \tag{293}$$

$$W=-\frac{1}{2}\epsilon_{\mu\nu\rho}P^{\mu}M^{\nu\rho}. \tag{7.294}$$

$$\varepsilon_{\pm}^{\mu}(-p)=-\varepsilon_{\mp}^{\mu}(p), \tag{A.295}$$

$$-\int\frac{\mathrm{d}^dl}{(2\pi)^d}\frac{1}{[l^2-m^2][(l+q)^2-m^2][p_1\cdot l-i0]}. \tag{296a}$$

$$\int\frac{\mathrm{d}^2\mathbf{l}}{(2\pi)^2}\,\mathbf{u}\cdot\frac{\partial}{\partial\mathbf{l}}\left(\frac{(\mathbf{u}\cdot\mathbf{l})^n}{[\mathbf{l}^2+m^2]^{\alpha}[(\mathbf{l}+\mathbf{q})^2+m^2]^{\beta}}\right)=0, \tag{297}$$

$$=-\frac{i}{2E\sqrt{1-\mathbf{u}^2}}\int\frac{\mathrm{d}^2\mathbf{l}}{(2\pi)^2}\frac{1}{[\mathbf{l}^2+m^2][(\mathbf{l}+\mathbf{q})^2+m^2]}, \tag{4.298}$$

$$\int\frac{\mathrm{d}\omega}{2\pi}\frac{\omega^n}{[\omega-\mathbf{u}\cdot\mathbf{l}+i0][\omega+\mathbf{u}\cdot\mathbf{l}-i0]}=\frac{i(-1)^{n-1}}{2}\left(\mathbf{u}\cdot\mathbf{l}-i0\right)^{n-1}. \tag{A.299}$$

$$\int\frac{\mathrm{d}^2\mathbf{q}}{(2\pi)^2}e^{i\mathbf{q}\cdot\mathbf{x}}\int\frac{\mathrm{d}^2\mathbf{l}}{(2\pi)^2}\frac{m^2}{[\mathbf{l}^2+m^2]^2[(\mathbf{l}+\mathbf{q})^2+m^2]}=\frac{mr}{8\pi^2}K_0(mr)K_1(mr). \tag{300a}$$

$$\int_0^{mr}\mathrm{d}r'\;r'\;K_0\left(mr'\right)^2I_0\left(mr'\right)=\frac{\pi}{3\sqrt{3}m^2}-\int_{mr}^{\infty}\mathrm{d}r'\;r'\;K_0\left(mr'\right)^2I_0\left(mr'\right), \tag{301}$$

$$a(\xi)=-Z_N\xi K_1(\xi)+\mathcal{O}\left(e^{-2\xi}\right), \tag{1.302}$$

$$\langle P(x)P^{\dagger}(x+R)\rangle_{N_t}=\sigma_0^{\frac{4-D}{2}}\frac{N_t}{\pi}\sum_{n=0}^{\infty}w_n\left(\frac{E_n}{2R}\right)^{\frac{1}{2}(D-3)}K_{\frac{1}{2}(D-3)}(E_nR) \tag{A.303}$$

$$E_n = \sigma_0 N_t \sqrt{1 + \frac{8\pi}{\sigma_0 N_t^2}\left[-\frac{1}{24}\left(D-2\right) + n\right]}. \tag{304a}$$

$$\langle P(x)P^\dagger(x+R)\rangle_{N_t} \sim \sigma_0^{\frac{4-D}{2}} N_t \left(\frac{E_0}{R}\right)^{\frac{1}{2}(D-3)} K_{\frac{1}{2}(D-3)}(E_0 R) \tag{305}$$

$$\langle P(x)P^\dagger(x+R)\rangle_{N_t} \sim N_t \sqrt{\sigma_0} K_0(E_0 R) \tag{5.306}$$

$$\frac{T_{c,NG}}{\sqrt{\sigma_0}} = \sqrt{\frac{3}{\pi(D-2)}} \tag{A.307}$$

$$\Delta(g) = \frac{1}{\sqrt{(g)}} \partial_a [\sqrt{(g)} g^{ab} \partial_b] \tag{308a}$$

$$\sigma(\beta) = \sigma_c t^{2\nu} \times \left(1 + a t^\theta + bt\right) , \tag{309}$$

$$\tilde{\gamma}_n = \gamma_n + (-1)^{(n+1)/2} \frac{1}{n 2^{3n-1}} \tag{2.310}$$

$$\ddot{q} = -V'(q) \tag{A.311}$$

$$E = \frac{1}{2}\dot{q}^2 + V(q) \tag{312a}$$

$$\partial_t^2 \phi(t,x) - \partial_x^2 \phi(t,x) = -V'(\phi) \tag{313}$$

$$i\hbar \frac{\partial \psi(t,\vec{x})}{\partial t} = \hat{H}\psi(t,\vec{x}). \tag{6.314}$$

$$\partial_t \psi(t \to t_c, \vec{x}_P) \to \infty. \tag{A.315}$$

$$\psi(t \to t_c, \vec{x}_P) \to \infty \quad \forall \vec{x}_P \tag{316a}$$

$$\psi(t > t_0) = e^{i\hat{H}(t-t_0)} \psi(t_0) \tag{317}$$

$$\psi(t,\vec{x}) = \int d^3x' K(\vec{x},t;\vec{x'},t_0)\psi(t_0,\vec{x'}) \tag{3.318}$$

$$\psi(t,\vec{x}) = \int d^3x' K(\vec{x},t;\vec{x'},t_c)\psi(t_c,\vec{x'}) \tag{A.319}$$

$$t(\tau_b) = t(\tau_d). \tag{320a}$$

$$\tau_c - \tau_b = N\epsilon, \quad \tau_d - \tau_c = N\epsilon, \quad \epsilon \to 0. \tag{321}$$

$$K_{II,III}(\tau_d, q_d; \tau_b, q_b) = \delta(q_b - q_d), \quad \tau_b \neq \tau_d. \tag{7.322}$$

$$S = \frac{1}{2}\int d^4x \left[\partial_\mu \phi \partial^\mu \phi - m^2 \phi^2\right] \tag{A.323}$$

$$\ddot{q}(t) + \omega^2 q(t) = 0. \tag{324a}$$

$$K(t_d, q_d; t_b, q_b) = \delta(q_d - q_b) \quad \text{at} \quad t_b = t_d \tag{325}$$

$$\tau_j = \frac{n_j^+ - n_j^-}{n_j} \tag{4.326}$$

$$\langle \tau_j \rangle = T_j, \qquad \mathrm{Var}(\tau_j) = \sigma_j^2 / n_j, \tag{A.327}$$

$$\langle \tau_j^2 \rangle = T_j^2 + \mathrm{Var}(\tau_j), \qquad \mathrm{Var}(\tau_j^2) = \frac{2(n_j - 1)(1 - T_j^2)[(2n_j - 3)T_j^2 + 1]}{n_j^3}. \tag{328a}$$

$$\langle S \rangle = \frac{(n-1)S_M^\infty + M}{n}, \quad \text{with} \quad \mathrm{Var}(S) = \sum_{i=1}^{M} \mathrm{Var}(\tau_i^2), \tag{329}$$

$$\rho = p|\psi\rangle\langle\psi| + (1-p)\rho_{\mathrm{n}}, \tag{1.330}$$

$$\mathrm{W} = \sigma_y \otimes \sigma_y \otimes \sigma_x - \sigma_x \otimes \sigma_x \otimes \sigma_x + 1, \tag{A.331}$$

$$\mathrm{p}_i(n_+) = \binom{n}{n_+} \left(\frac{1+T_i}{2}\right)^{n_+} \left(\frac{1-T_i}{2}\right)^{n_-}, \tag{332a}$$

$$\mathrm{P}(\tau_i) = \mathrm{p}_i\left(\frac{n}{2}(1+T_i)\right). \tag{333}$$

$$\mathrm{P}(\tau_i^2) = \mathrm{P}(\tau_i) + \mathrm{P}(-\tau_i) \quad \text{for} \quad \tau_i \neq 0, \tag{5.334}$$

$$\mathrm{P}(S) = \sum_{\tau_1^2 + \cdots + \tau_M^2 = S} \mathrm{P}(\tau_1^2) \cdots \mathrm{P}(\tau_M^2). \tag{A.335}$$

$$\mathrm{P}(Q^\infty \in \mathcal{O}_{\mathrm{acc}} | \mathrm{sep}) = 0. \tag{336a}$$

$$\sum_{Q \in \mathcal{O}_{\mathrm{rej}}} \mathrm{P}(Q|\mathrm{sep}) \geq q_{\mathrm{F}}. \tag{337}$$

$$r = 1 - e_2 = \sum_{Q \in \mathcal{O}_{\mathrm{acc}}} \mathrm{P}(Q|\mathrm{ent}). \tag{2.338}$$

$$\mathrm{P}(\mathrm{ent}|Q) \geq q_{\mathrm{B}}, \quad \text{for all} \quad Q \in \mathcal{O}_{\mathrm{acc}}, \tag{A.339}$$

$$L_Q = \begin{cases} \beta\, \mathrm{P}(\mathrm{sep}|Q) & \text{if } Q \in \mathcal{O}_{\mathrm{acc}} \\ (1-\beta)\, \mathrm{P}(\mathrm{ent}|Q) & \text{if } Q \in \mathcal{O}_{\mathrm{rej}}, \end{cases} \tag{340a}$$

$$T_{\mu_1 \ldots \mu_N} = \langle R_{\mu_1} \ldots R_{\mu_N} \rangle = \mathrm{Tr}(\rho\, \sigma_{\mu_1} \otimes \cdots \otimes \sigma_{\mu_N}), \tag{341}$$

$$\langle E_{\mathrm{W}} \rangle = E_{\mathrm{W}}^\infty, \quad \text{and} \quad \mathrm{Var}(E_{\mathrm{W}}) = \sum_j \alpha_j^2 \mathrm{Var}(\tau_j). \tag{6.342}$$

$$\mathrm{P}(p) = \begin{cases} \mathcal{N}\mathrm{e}^{-\frac{(p-\bar{p})^2}{2(\Delta p)^2}} & p \in [p_{\min}, 1] \\ 0 & \text{else,} \end{cases} \tag{A.343}$$

$$\mathrm{W} = \sum_j \alpha_j \sigma_j. \tag{344a}$$

$$E_{\mathrm{W}} = \mathrm{Tr}(\mathrm{W}\rho) = \sum_j \alpha_j T_j. \tag{345}$$

$$S \equiv \sum_{j=1}^{3^N} T_j^2, \tag{3.346}$$

$$\begin{aligned} \mathrm{P}(Q|\mathrm{sep}) &= \sum_j \mathrm{P}(Q|\rho_j)\mathrm{P}(\rho_j|\mathrm{sep}), \\ \mathrm{P}(Q|\mathrm{ent}) &= \sum_j \mathrm{P}(Q|\rho_j)\mathrm{P}(\rho_j|\mathrm{ent}). \end{aligned} \tag{A.347}$$

$$\sum_{Q\in\mathcal{O}_{\mathrm{acc}}} \mathrm{P}(Q|\mathrm{sep}) \leq \sum_{Q\in\mathcal{O}_{\mathrm{acc}}} \mathrm{G}_{\mathrm{acc}}(Q|\mathrm{sep}) \leq 1 - q_{\mathrm{F}}. \tag{348a}$$

$$\delta\Phi \leq \frac{1}{2\sqrt{\frac{t}{T_r}}\left(\frac{\partial P_{|1\rangle}}{\partial \Phi}\right)} = \frac{A}{\sqrt{t}}, \tag{349}$$

$$P_{|1\rangle}(\Phi, \tau) = \frac{1}{2} + \frac{1}{2} e^{-A\tau - B^2\tau^2} \cos(\Delta\omega(\Phi)\tau), \tag{7.350}$$

$$P_{|1\rangle} = \frac{1}{2} + \frac{1}{2} \frac{e^{\frac{hf_q}{k_B T}} - 1}{e^{\frac{hf_q}{k_B T}} + 1} e^{-A\tau - B^2\tau^2} \cos(\Delta\omega\tau), \tag{A.351}$$

$$P_{|10..0\rangle} = \frac{1}{2} + \frac{1}{2} \frac{e^{\frac{hf_q}{k_B T}} - 1}{e^{\frac{hf_q}{k_B T}} + 1} e^{-N(A\tau + B^2\tau^2)} \cos(N\Delta\omega\tau). \tag{352a}$$

$$\overline{\tau_l} = \frac{1}{F} \sum_{j=1}^{F} \frac{1}{M} \sum_{k=1}^{M} \tau_{j,k,l}, \quad \tau_{j,k,l} = \sum_{i=1}^{l} \tau_i^{(j,k)} n_i^{(j,k)}. \tag{353}$$

$$\overline{\left(\frac{\delta\Phi}{\Phi_0}\right)_l} = \sqrt{\frac{1}{\Phi_0^2 F} \sum_{j=1}^{F} \frac{1}{M-1} \sum_{k=1}^{M} (\hat{\Phi}_{jkl} - \Phi_j)^2}. \tag{4.354}$$

$$\vec{B}(x, y) = \frac{\mu_0}{4\pi} \int_l \frac{I d\vec{l} \times \vec{r}'}{|\vec{r}'|^3} \tag{A.355}$$

$$\Phi = \int_{x_1}^{x_2} \int_{y_1}^{y_2} B(x, y) dx dy. \tag{356a}$$

$$S_x[\omega] = \frac{2k_B T\gamma}{M[(\omega_m^2 - \omega^2)^2 + \gamma^2\omega^2]}\,. \tag{357}$$

$$S_x[\omega] = \frac{2k_B T\gamma}{M[(\omega_m^2 - \omega^2)^2 + \gamma^2(1+g)^2\omega^2]}\,. \tag{1.358}$$

$$y_{\rm out} = \sqrt{\kappa} y_c - y_{\rm c,in} \tag{A.359}$$

$$A[\omega] = A_0 \exp\left(-i\phi\frac{\omega}{\omega_m}\right)\,. \tag{360a}$$

$$\chi_{\rm fb}[\omega] = \frac{1}{-i\omega\gamma_{\rm eff} + \omega_{\rm eff}^2 - \omega^2}\,. \tag{361}$$

$$\phi_m = \tan^{-1}\left(-\frac{\kappa}{2\omega_m}\right) + \pi \tag{5.362}$$

$$\gamma_{\rm fb} = \frac{4GA_0}{\sqrt{\kappa^2 + 4\omega_m^2}}\,. \tag{A.363}$$

$$n_T = \frac{\gamma}{\gamma_{\rm eff}}\left(n_m^T + \frac{1}{2}\right)\,. \tag{364a}$$

$$n_{\rm qba} = C_{\rm eff}\frac{\kappa^2}{\kappa^2 + 4\omega_m^2}\,. \tag{365}$$

$$C_{\rm eff} = \frac{4G^2}{\kappa\gamma_{\rm eff}}\,. \tag{2.366}$$

$$n_m + \frac{1}{2} = n_T + n_{\rm qba} + n_{\rm fb}\,, \tag{A.367}$$

$$n_{m,\rm min} + \frac{1}{2} = \frac{1}{2}\sqrt{1 + 2n_{\rm add}}\,. \tag{368a}$$

$$S_{\rm out}[\omega] = \langle a_{\rm out}^\dagger[\omega] a_{\rm out}[-\omega]\rangle + \frac{1}{2}\,. \tag{369}$$

$$\gamma_{\rm opt} = \frac{4G^2}{\kappa}\frac{1}{1 + \left(\frac{\kappa}{4\omega_m}\right)^2}\,. \tag{6.370}$$

$$\gamma_{\rm eff} = \gamma - \gamma_{\rm opt} + \frac{4GA_0\left[(\kappa^2 + 8\omega_m^2)\sin\phi - 2\kappa\omega_m\cos\phi\right]}{\kappa^3 + 16\kappa\omega_m^2}\,. \tag{A.371}$$

$$\begin{aligned} X_f &= \chi_{\rm fb}\,, \\ X_{\rm inj} &= -\frac{4G\omega_m\sqrt{\kappa}}{\kappa - 2i\omega}\chi_{\rm fb}\,, \end{aligned} \tag{372a}$$

$$x[\omega] \equiv \frac{b[\omega] + b^\dagger[\omega]}{\sqrt{2}}\,, \tag{373}$$

$$x_{\varphi}[\omega] \equiv \frac{b[\omega]e^{i\varphi} + b^{\dagger}[\omega]e^{-i\varphi}}{\sqrt{2}} \tag{3.374}$$

$$s[\omega] = \frac{2\mathcal{G}\sqrt{\kappa}G}{\sqrt{\kappa^2/4 + \omega_m^2}}\, x_{\phi_0+\phi_\tau}[\omega]\,. \tag{A.375}$$

$$A_0 = \frac{g_0\sqrt{\kappa}\alpha_0^2\mathcal{G}}{\sqrt{\kappa^2/4 + \Delta_f^2}} D \tag{376a}$$

$$\gamma_{\rm opt} = \mathcal{P}P\,, \tag{377}$$

$$G_{\rm rsb} = \frac{1}{2}\sqrt{\mathcal{P}P\kappa\left[1 + \left(\frac{\kappa}{4\omega_m}\right)^2\right]}\,. \tag{7.378}$$

$$G = \frac{|\chi_c(\Delta)|}{|\chi_c(-\omega_m)|} G_{\rm rsb}\,, \tag{A.379}$$

$$\gamma_{\rm fb} = \mathcal{L}g\,. \tag{380a}$$

$$A_0 = \frac{\mathcal{L}\mathcal{G}\sqrt{\kappa^2 + 4\omega_m^2}}{4G}\,. \tag{381}$$

$$\gamma_{\rm eff} = \gamma - \gamma_{\rm opt} + \gamma_{\rm fb,bsb}\,, \tag{4.382}$$

$$\gamma_{\rm fb,bsb} = \frac{4A_0G\sqrt{\kappa^4 + 20\kappa^2\omega_m^2 + 64\omega_m^4}}{\kappa^3 + 16\kappa\omega_m^2}\,. \tag{A.383}$$

$$-\sqrt{2}Ga^{\dagger}a = -\sqrt{2}G^2\kappa^2\mathcal{G}\mathcal{A}|\chi_c[\omega_m]\chi_c[\omega_f]|\left[|\chi_c[\omega_f - \omega_m]|x_{\Psi-}(t) \;+\; |\chi_c[\omega_f + \omega_m]|x_{\Psi+}(t)\right] \tag{384a}$$

$$\begin{aligned} \Psi- &= \phi_\tau + \phi_0 - \phi_- - \phi_1 \\ \Psi_+ &= \phi_\tau + \phi_0 + \phi_+ + \phi_1 \end{aligned} \tag{385}$$

$$\hat{H}(x_1, x_2, \ldots x_N) = \sum_{i=1}^{N} \hat{h}(x_i) + \sum_{i<j=1}^{N} \hat{W}(x_i - x_j) \tag{1.386}$$

$$|\psi(t)\rangle = \sum_{\bar{n}} C_{\bar{n}}(t)|\bar{n}; t\rangle, \tag{A.387}$$

$$|\bar{n}; t\rangle = \prod_{i=1}^{M}\left(\frac{\left(b_i^{\dagger}(t)\right)^{n_i}}{\sqrt{n_i!}}\right)|vac\rangle. \tag{388a}$$

$$\hat{W}(x_i - x_j) = \lambda\delta(x_i - x_j) \tag{389}$$

$$\rho^{(1)}(x_1'|x_1;t) = N \int dx_2\, dx_3 ... dx_N\, \psi^*(x_1', x_2, \ldots, x_N; t) \psi(x_1, x_2, \ldots, x_N; t). \tag{5.390}$$

$$\rho^{(1)}(x_1'|x_1;t) = \sum_{kq} \rho_{kq} \phi_k^*(x_1';t)\phi_q(x_1;t) \tag{A.391}$$

$$\rho(x;t) = \rho^{(1)}(x_1' = x|x_1 = x;t) \tag{392a}$$

$$\rho^{(p)}(x_1, \ldots, x_p;t) = \langle\psi(t)|\hat{\psi}^\dagger(x_1)\ldots\hat{\psi}^\dagger(x_p)\hat{\psi}(x_p) \ldots\hat{\psi}(x_1)|\psi(t)\rangle \tag{393}$$

$$g^{(p)}(x_1, \ldots, x_p;t) = \frac{\rho^{(p)}(x_1, \ldots, x_p;t)}{\Pi_{i=1}^p |\rho^{(1)}(x_i)|} \tag{2.394}$$

$$\Lambda(t) = 1 - n_1(t) \tag{A.395}$$

$$H = \sum_{\langle xy\rangle} \left[-\frac{t}{2}(c_x^\dagger c_y + c_y^\dagger c_x) + V(n_x - \frac{1}{2})(n_y - \frac{1}{2}) \right] \tag{396a}$$

$$H_{N_f}^{(g)} = -\sum_{\langle xy\rangle} \prod_{f=1}^{N_f} \left(H_{\langle xy\rangle,f}^{(g)} + H_{\langle xy\rangle,f}^{(g),\mathrm{des}} \right) \tag{397}$$

$$H_{N_f=2}^{U(1)} \rightarrow H_{N_f=2}^{U(1)} + J \sum_{\langle xy\rangle} s_{xy,1}^3 s_{xy,2}^3, \tag{6.398}$$

$$\begin{aligned} w_A &= 1 \\ w_D &= \exp(\epsilon t)\sinh \epsilon t \\ w_A + w_D &= \exp(\epsilon t)\cosh \epsilon t, \end{aligned} \tag{A.399}$$

$$1 + \left(\frac{1}{2} e^{2\epsilon t} - \frac{1}{2} \right) = \left(\frac{1}{2} e^{2\epsilon t} + \frac{1}{2} \right) \tag{400a}$$

$$\begin{aligned} &1 + \left(\frac{1}{2^{N_f}} e^{\epsilon (2t)^{N_f}} - \frac{1}{2^{N_f}} \right) \\ &= \left(\frac{1}{2^{N_f}} e^{\epsilon (2t)^{N_f}} + \left(1 - \frac{1}{2^{N_f}} \right) \right), \end{aligned} \tag{401}$$

$$\begin{aligned} H_e^s &= \sum_x s_{2x,2x+1,1}^3 s_{2x,2x+1,2}^3, \\ H_o^s &= \sum_x s_{2x+1,2x+2,1}^3 s_{2x+1,2x+2,2}^3. \end{aligned} \tag{3.402}$$

$$e^{\tau_1 J/4} e^{\tau_2 J/4} \tag{A.403}$$

$$e^{-\tau_1 J/4} e^{-\tau_2 J/4} \tag{404a}$$

$$V \propto X_0 + X_0^{\dagger}. \tag{405}$$

$$\mathcal{R}^{(3)} \propto \langle [[[V\left(t_1+t_2+t_3\right), V\left(t_1+t_2\right)], V\left(t_1\right)], V] \rangle, \tag{7.406}$$

$$R_2 = \langle VV\left(t_1+t_2\right) V\left(t_1+t_2+t_3\right) V\left(t_1\right) \rangle, \tag{A.407}$$

$$R_3 = \langle VV\left(t_1\right) V\left(t_1+t_2+t_3\right) V\left(t_1+t_2\right) \rangle, \tag{408a}$$

$$R_1^* = -\langle VV\left(t_1+t_2+t_3\right) V\left(t_1+t_2\right) V\left(t_1\right) \rangle, \tag{409}$$

$$R_1^* = -\langle \mathrm{FS}| X_0 e^{i\mathcal{H}_1(t_1+t_2+t_3)} X_0 e^{-i\mathcal{H}_2 t_3} X_0^{\dagger} e^{-i\mathcal{H}_1 t_2} X_0^{\dagger} |\mathrm{FS}\rangle, \tag{4.410}$$

$$R_1^* = -\sum_{nm\eta} \phi_0^{(n)} \Phi^{(n,\eta)} \Phi^{(m,\eta)*} \phi_0^{(m)*} e^{i\mathcal{E}_n t_1} e^{i[\mathcal{E}_n - \mathcal{E}_m] t_2} e^{i[\mathcal{E}_n - \mathcal{E}_\eta] t_3}, \tag{A.411}$$

$$\int\int d\omega_1 d\omega_3 \mathcal{S}_{\mathrm{ESA}} = -\int\int d\omega_1 d\omega_3 \mathcal{S}_{\mathrm{ESE}} - \int\int d\omega_1 d\omega_3 \mathcal{S}_{\mathrm{GSB}}. \tag{412a}$$

$$|n\rangle = \phi_0^{(n)} |0\rangle_1 + \sum_{k_p k_h} \phi_{k_p k_h}^{(n)} |k_p k_h\rangle_1, \tag{413}$$

$$|\eta\rangle = \sum_k{}' \Phi_k^{(\eta)} |k\rangle_2 + \sum_k{}' \sum_{k_p k_h} \Phi_{k k_p k_h}^{(\eta)} |k k_p k_h\rangle_2, \tag{1.414}$$

$$|k k_p k_h\rangle_2 = C_{k,k_p,k_h} X_{-k}^{\dagger} X_{k-k_p+k_h}^{\dagger} c_{k_p}^{\dagger} c_{k_h} |\mathrm{FS}\rangle \tag{A.415}$$

$$\boldsymbol{r} = a(t)[\boldsymbol{q} - D_+(t) \boldsymbol{\nabla} \Phi], \tag{416a}$$

$$\Phi(\boldsymbol{q}) = \Phi(\boldsymbol{q}_{cm}) + \Delta q^i \left.\frac{\partial \Phi}{\partial q^i}\right|_{q_{cm}} + \Delta q^i \Delta q^j \left.\frac{\partial^2 \Phi}{\partial q_i \partial q_j}\right|_{q_{cm}} \tag{417}$$

$$L_i \approx -t_m \epsilon_{ijk} I_{jl} T_{lk} \tag{5.418}$$

$$\tau_i = \epsilon_{ijk} I_{jl} \partial_l \partial_k \Phi. \tag{A.419}$$

$$L_i = \epsilon_{ijk} I_{jl} \mathcal{M}_{lk} \tag{420a}$$

$$\mathcal{M}_{ij} = \frac{2GM_2}{b^2 v} (\hat{b}_i \hat{b}_j - \hat{c}_i \hat{c}_j) \tag{421}$$

$$\lambda = \frac{L\sqrt{|E|}}{GM^{5/2}}, \tag{2.422}$$

$$\lambda_{TLT} \sim \left(\frac{GM_2}{v^2 b}\right)^{1/2} \left(\frac{R_1}{b}\right)^{3/2} \tag{A.423}$$

$$i\frac{d}{dr} \begin{bmatrix} \nu_L \\ \nu_R \end{bmatrix} = \begin{bmatrix} H_L + (\dot{\phi}/2) I & \mu B(r) \\ \mu^{\dagger} B(r) & H_R - (\dot{\phi}/2) I \end{bmatrix} \begin{bmatrix} \nu_L \\ \nu_R \end{bmatrix}, \tag{424a}$$

$$V_\alpha + \dot{\phi} = 0. \tag{425}$$

$$\gamma_\alpha = \frac{2(2\mu_\nu B)^2}{|\dot{V}_\alpha + \ddot{\phi}|}. \tag{6.426}$$

$$\Phi_e^{NO} = |U_{e3}|^2\Phi_{ie} + (1-|U_{e3}|^2)\Phi_{ix} \approx 0.02\Phi_{ie} + 0.98\Phi_{ix}, \tag{A.427}$$

$$\Phi_e^{IO} = |U_{e2}|^2\Phi_{ie} + (1-|U_{e2}|^2)\Phi_{ix} \approx 0.3\Phi_{ie} + 0.7\Phi_{ix}. \tag{428a}$$

$$\Phi_{\bar{e}}^{NO} = |U_{e1}|^2\Phi_{i\bar{e}} + (1-|U_{e1}|^2)\Phi_{ix} \approx 0.7\Phi_{i\bar{e}} + 0.3\Phi_{ix}, \tag{429}$$

$$\Phi_{\bar{e}}^{IO} = |U_{e3}|^2\Phi_{i\bar{e}} + (1-|U_{e3}|^2)\Phi_{ix} \approx 0.02\Phi_{i\bar{e}} + 0.98\Phi_{ix}. \tag{3.430}$$

$$\Phi_{i\alpha} \to e^{-\frac{\pi}{2}\gamma_\alpha}\Phi_{i\alpha}, \tag{A.431}$$

$$\frac{dN_{\nu_\alpha}}{dE_r} = \frac{N_{\rm t}}{4\pi R^2}\int dE_t \Phi_\alpha\left(E_t\right)\sigma_\alpha\left(E_t\right)W\left(E_r,E_t\right) \tag{432a}$$

$$\chi^2 = \min_{\xi}\sum_{i=1}^{n} 2\left[(1+\xi)F_i - D_i + D_i\ln\left(\frac{D_i}{(1+\xi)F_i}\right)\right], \tag{433}$$

$$\tilde{\phi} \;\equiv\; A(\chi)\phi\,, \qquad \tilde{\psi} \;\equiv\; A^{3/2}(\chi)\psi\,, \tag{7.434}$$

$$A^2(\chi) \;=\; 1 \;+\; b\,\frac{\chi}{M} \;+\; c\,\frac{\chi^2}{M^2} \;+\; \mathcal{O}\!\left(\frac{\chi^3}{M^3}\right), \tag{A.435}$$

$$V(\chi) \;=\; \frac{1}{2}\,\mu_\chi^2\,\chi^2\,. \tag{436a}$$

$$\tilde{\chi} \;\equiv\; \left(1 \;+\; \frac{b^2\tilde{\phi}^2}{4M^2}\right)^{1/2}\chi\,, \tag{437}$$

$$V(\tilde{\chi}) = \frac{1}{2}\mu_\chi^2\tilde{\chi}^2 + \frac{3}{2}\,\frac{\mu^4}{\lambda}\left(1 + 2b\,\frac{\tilde{\chi}}{M} + 2c\,\frac{\tilde{\chi}^2}{M^2} + b^2\,\frac{\tilde{\chi}^2}{M^2}\right), \tag{4.438}$$

$$\frac{\mu^2\tilde{\phi}_m^4}{v^2} - \tilde{\phi}_m^2\left[\mu^2\left(1 + \frac{b\tilde{\chi}_m}{M} + \frac{c\tilde{\chi}_m^2}{M^2}\right) + \frac{b\mu_\chi^2\tilde{\chi}_m^2}{4M^2}\right] + \rho_\psi = 0\,, \tag{A.439}$$

$$\tilde{\chi}_m\left[\frac{\mu^2v^2}{M^2}\left(c - \frac{c\tilde{\phi}_m^2}{v^2} + \frac{b^2}{2}\right) + \mu_\chi^2\left(1 - \frac{b^2\tilde{\phi}_m^2}{4M^2}\right)\right] = \frac{b\mu^2v^2}{2M}\left(1 - \frac{\tilde{\phi}_m^2}{v^2}\right), \tag{440a}$$

$$\frac{\tilde{\chi}_m}{M} = -\frac{b\rho_\psi}{2\mu_\chi^2M^2 + (2c-b^2)\rho_\psi}\,. \tag{441}$$

$$\theta \approx -\frac{v}{2M}\left[b + \frac{(4c-b^2)\tilde{\chi}_m}{2M} + \frac{\rho_\psi}{\mu^2v^2}\left(b + \frac{2c\tilde{\chi}_m}{M} - \frac{5b^2\tilde{\chi}_m}{2M}\right)\right]. \tag{1.442}$$

$$\beta(\rho_\psi) = \frac{2b\mu_\chi^2 M^2}{2\mu_\chi^2 M^2 + (2c-b^2)\rho_\psi} + \mathcal{O}\left(\frac{\rho_\psi}{\mu^2 v^2}\right) \ . \tag{A.443}$$

$$F_s = -\frac{\beta(\rho_\psi)}{M}\nabla s \ . \tag{444a}$$

$$\frac{\rho_\psi}{\mathrm{gcm}^{-3}} \gtrsim \frac{0.46}{2c-b^2}\left(\frac{\mu_\chi}{\mathrm{eV}}\right)^2\left(\frac{M}{\mathrm{GeV}}\right)^2 \ , \tag{445}$$

$$\beta(\rho_\psi) \approx \frac{2b\mu_\chi^2 M^2}{(2c-b^2)\rho_\psi} \ , \tag{5.446}$$

$$\frac{1}{(2c-b^2)^{1/2}}\left(\frac{\mu_\chi}{\mathrm{eV}}\right)\left(\frac{M}{\mathrm{GeV}}\right) \lesssim 2.1\times 10^{-13} \ . \tag{A.447}$$

$$\frac{1}{(2c-b^2)^{1/2}}\left(\frac{\mu_\chi}{\mathrm{eV}}\right)\left(\frac{M}{\mathrm{GeV}}\right) < 5.1 \ . \tag{448a}$$

$$\frac{M_c}{R} \gg \frac{|b^2-2c|M^2}{b^2} \ . \tag{449}$$

$$2.1\times 10^{-13} \lesssim \frac{1}{(2c-b)^{1/2}}\left(\frac{\mu_\chi}{\mathrm{eV}}\right)\left(\frac{M}{\mathrm{GeV}}\right) \lesssim 5.1 \ . \tag{2.450}$$

$$\frac{M}{\mathrm{GeV}} \ll \frac{(3\times 10^{14})b}{|b^2-2c|^{1/2}} \ . \tag{A.451}$$

$$\begin{aligned} \alpha_{hs} &= \frac{b\mu^2}{2M} \ , \\ \lambda_{hs} &= \frac{c\mu^2}{2M^2} + \frac{b^2\mu_\chi^2}{8M^2} \ . \end{aligned} \tag{452a}$$

$$\begin{aligned} h &= \delta\tilde{\phi}\cos\theta + \delta\tilde{\chi}\sin\theta \ , \\ s &= \delta\tilde{\chi}\cos\theta - \delta\tilde{\phi}\sin\theta \ . \end{aligned} \tag{453}$$

$$d(y,\vec{C}) - d(y,\vec{Y}) \leq cost(C_j,\vec{C}) - cost(C_j,\vec{Y}) \tag{6.454}$$

$$d(C_i, C_{i+1}) > \Big(\frac{\gamma^2+1}{\gamma+1}\frac{1}{c} - 1\Big)D(C_i) \tag{A.455}$$

$$d(C_i, C_{i+1}) > \Big(\frac{c-1}{c}(\gamma-1)\Big)D(C_i) \tag{456a}$$

$$d(C_i, C_{i+1}) > \max\Big\{\frac{\gamma^2+1}{\gamma+1}\frac{1}{c} - 1, \frac{c-1}{c}(\gamma-1)\Big\}D(C_i) \tag{457}$$

$$c = \frac{2\gamma^2}{\gamma^2+\gamma} \tag{3.458}$$

$$w > \frac{\gamma^2 - \gamma}{\gamma + 1}\frac{y}{2} + (\gamma - 2)\frac{z}{2} \tag{A.459}$$

$$\begin{aligned} D(C_i) = d(x_{i,l}, x_{i,r}) = d(c_i, x_{i,r}) < \frac{1}{(\gamma - 1)} d(x_{i,r}, x_{i+1,r}) = \frac{1}{(\gamma - 1)} d(C_i, C_{i+1}) \Rightarrow \\ d(C_i, C_{i+1}) > (\gamma - 1) D(C_i) \end{aligned} \tag{460a}$$

$$\frac{\gamma^2 + 1}{\gamma + 1}\frac{1}{c} - 1 = \frac{c - 1}{c}(\gamma - 1) \tag{461}$$

$$d(C_i, C_{i+1}) > \Big(\frac{\gamma^2 + 1}{2\gamma} - 1\Big) D(C_i) = \frac{(\gamma - 1)^2}{2\gamma} D(C_i) \tag{7.462}$$

$$cost(\vec{x}, M(\vec{x})) = \sum_{i=1}^{n} \mathbb{E}_{\vec{c} \sim M(\vec{x})}(d(x_i, \vec{c}))\,. \tag{A.463}$$

$$\gamma - 1 \geq \frac{\gamma^2 + 1}{2\gamma} - 1 \tag{464a}$$

$$D(C_i) = d(x_{i,l}, x_{i,r}) = d(c_i, x_{i,r}) < \frac{1}{(\gamma - 1)} d(x_{i,r}, x_{i-1,r}) = \frac{1}{(\gamma - 1)} d(C_i, C_{i+1}) < d(C_i, C_{i+1}) \tag{465}$$

$$D(C_i) + d(C_i, C_{i+1}) = d(x_{i,l}, c_i) + d(c_i, x_{i+1,l}) = d(x_{i,l}, x_{i+1,l}) > (\gamma - 1) d(x_{i,l}, c_i) \geq 2D(C_i)\,, \tag{4.466}$$

$$\begin{aligned} \frac{\gamma^2 - 3\gamma - 2}{\gamma + 1}\frac{(\gamma - 2)}{2} + \frac{\gamma - 2}{2} \geq 1 \Rightarrow \\ \gamma^3 - 4\gamma^2 + \gamma \geq 0 \end{aligned} \tag{A.467}$$

$$\tilde{Q} := \begin{bmatrix} Q & \mathbf{u} \\ \mathbf{v}^T & \tau \end{bmatrix}. \tag{468a}$$

$$\text{T-value}(\mathbf{x}_i) = \frac{\mu_{i,\diamond} - \mu_{i,\oslash}}{\sqrt{\big(\sigma^2_{i,\diamond} + \sigma^2_{i,\oslash}\big)/M}}. \tag{469}$$

$$\text{MV}_{\xi}(\mathbf{x}) := \arg\max_{y \in \mathcal{Y}} \mathbb{P}_{\theta \sim \xi}\left[h_\theta(\mathbf{x}) = y\right]. \tag{1.470}$$

$$\left|\hat{\theta} - \frac{\Delta(\mu_+ - \mu_-)}{2} - \frac{(\mu_+ + \mu_-)}{2}\right| \leq \delta \tag{A.471}$$

$$P(E) \geq 1 - 2\exp\Big\{-\frac{2\delta^2}{9\sigma^2[1/\tilde{n}_+ + 1/\tilde{n}_-]}\Big\} - \frac{9\texttt{Var}(I_i^+)}{4\delta^2} - \frac{9\texttt{Var}(I_i^-)}{4\delta^2}. \tag{472a}$$

$$\text{minimize}_{Q \in \mathbb{R}^{N \times K}} \left\langle Q, C \right\rangle \tag{473}$$

$$\mathcal{L}(Q) = \text{minimize}_{Q \in \mathbb{R}^{N \times K}} \left\langle Q, C \right\rangle - \epsilon \mathbb{H}(Q) \tag{5.474}$$

$$\mathbf{a} = \exp\left(\frac{\lambda^{(1)}}{\epsilon} - 1/2\right) \tag{A.475}$$

$$\mathbf{b} = \exp\left(\frac{\lambda^{(2)}}{\epsilon} - 1/2\right). \tag{476a}$$

$$Q^* = \operatorname{diag}(\mathbf{a}^{(J)}) \exp\left(-\frac{C_{i,k}}{\epsilon}\right) \operatorname{diag}(\mathbf{b}^{(J)}). \tag{477}$$

$$\Delta_{\mathcal{D}}(\theta) \leq \sqrt{\frac{\gamma_{\mathcal{F}}(\xi, \pi) + \ln(N/\delta)}{2(N-1)}} \tag{2.478}$$

$$\mu_{i,\diamond} = \frac{1}{M} \sum_{m=1}^{M} p_m(y = \diamond \mid \mathbf{x}_i) \tag{A.479}$$

$$\mu_{i,\oslash} = \frac{1}{M} \sum_{m=1}^{M} p_m(y = \oslash \mid \mathbf{x}_i) \tag{480a}$$

$$\sigma_{i,\diamond}^2 = \frac{1}{M} \sum_{m=1}^{M} \Big(p_m(y = \diamond \mid \mathbf{x}_i) - \mu_{i,\diamond}\Big)^2 \tag{481}$$

$$\sigma_{i,\oslash}^2 = \frac{1}{M} \sum_{m=1}^{M} \Big(p_m(y = \oslash \mid \mathbf{x}_i) - \mu_{i,\oslash}\Big)^2. \tag{6.482}$$

$$\sum_{k=1}^{K} \Big|\hat{\theta}_k - \mu_k\Big| \leq \delta \tag{A.483}$$

$$\sum_{k=1}^{K} \Big|\hat{\theta}_k - \mu_k\Big| = \sum_{k=1}^{K} \Big|\hat{\theta}_k - \mu_k\Big| - \sum_{k=1}^{K} \mathbb{E}(I^k) \left[\mu_k - \mu_{\setminus k}\right]. \tag{484a}$$

$$P(E) \geq 1 - 2\sum_{k=1}^{K}\sum_{j=1}^{d} \exp\Big\{-\frac{\delta^2 \tilde{n}_k}{8\sigma_j^2}\Big\} - \sum_{k=1}^{K} \frac{4\mathtt{Var}(I^k)}{\delta^2} \left|\mu_k - \mu_{\setminus k}\right|. \tag{485}$$

$$\mathbb{H}\left[p(y \mid \mathbf{x}, D)\right] \sim \mathbb{H}\left[\frac{1}{M} \sum_{m=1}^{M} \underbrace{p_m\Big(y = 1, ..., K \mid \mathbf{x}\Big)}_{\in \mathbb{R}^K}\right] \tag{3.486}$$

$$L(\mathbf{x}; \tilde{q}) := \sum_{j=1}^{M} q_j P(y = k \mid \mathbf{x}, \theta_j), \tag{A.487}$$

$$\arg\max_{\mathbf{x} \in \mathcal{D}_u} L(\mathbf{x}; \tilde{q}). \tag{488a}$$

$$\arg\max_{\mathbf{x} \in \mathcal{D}_u} \min_{\tilde{q} \in B(\varepsilon)} L(\mathbf{x}; \tilde{q}), \tag{489}$$

$$\max_{\tilde{q}\in B(\varepsilon)} L(\mathbf{x};\tilde{q}) = \frac{1}{M}\sum_{m=1}^{M} p_{mk} - \sqrt{\varepsilon \sum_{m=1}^{M}\left(\frac{1}{M}p_{mk} - p_{mk}\right)^2}. \tag{7.490}$$

$$C := \begin{bmatrix} -\log \bar{P} & \mathbf{0}_N \\ \mathbf{0}_K^T & 0 \end{bmatrix} \tag{A.491}$$

$$L(\mathbf{x}_k, t_k) - L(\mathbf{x}_k, t_k - \Delta t_k) = p_k\, C, \tag{492a}$$

$$e_k \doteq (\mathbf{x}_k, t_k, p_k) \quad \overset{\mathbf{W}}{\mapsto} \quad e'_k \doteq (\mathbf{x}'_k, t_{\text{ref}}, p_k), \tag{493}$$

$$I(\mathbf{x};\boldsymbol{\theta}) \doteq \sum_{k=1}^{N_e} b_k\, \delta(\mathbf{x} - \mathbf{x}'_k(\boldsymbol{\theta})). \tag{4.494}$$

$$G(\boldsymbol{\theta}) \equiv \mathrm{Var}\big(I(\mathbf{x};\boldsymbol{\theta})\big) \doteq \frac{1}{|\Omega|}\int_{\Omega} (I(\mathbf{x};\boldsymbol{\theta}) - \mu_I)^2 d\mathbf{x}, \tag{A.495}$$

$$\mathbf{x}'_k = (1 - \tilde{t}_k h_z)\, \mathbf{x}_k, \tag{496a}$$

$$\mathcal{D}(\mathcal{E},\boldsymbol{\theta}) \doteq \{\nabla \cdot \mathbf{f}_k\}_{k=1}^{N_e}, \tag{497}$$

$$\mathrm{DIWE}(\mathbf{x};\mathcal{E},\boldsymbol{\theta}) \doteq \frac{1}{N_e(\mathbf{x})}\sum_{k} (\nabla \cdot \mathbf{f}_k)\, \delta(\mathbf{x} - \mathbf{x}'_k), \tag{1.498}$$

$$R_{\mathrm{DIWE}}(\mathcal{E},\boldsymbol{\theta}) \doteq \frac{1}{|\Omega|}\int_{\Omega} \mathrm{DIWE}(\mathbf{x};\mathcal{E},\boldsymbol{\theta}) d\mathbf{x}. \tag{A.499}$$

$$\mathcal{A}(\mathcal{E},\boldsymbol{\theta}) \doteq \big\{|\det(\mathtt{J}(e_k))|\big\}_{k=1}^{N_e}, \tag{500a}$$

$$\mathrm{IWA}(\mathbf{x}) \doteq 1 + \frac{1}{N_e(\mathbf{x})}\sum_{k=1}^{N_e} \big(|\det(\mathtt{J}(e_k))| - 1\big)\, \delta(\mathbf{x} - \mathbf{x}'_k). \tag{501}$$

$$R_{\mathrm{IWA}}(\mathcal{E},\boldsymbol{\theta}) \doteq \frac{1}{|\Omega|}\int_{\Omega} \mathrm{IWA}(\mathbf{x};\mathcal{E},\boldsymbol{\theta}) d\mathbf{x}. \tag{5.502}$$

$$\mathbf{x}^{h\prime} \sim \mathtt{R}(t\boldsymbol{\omega})\, \mathbf{x}^h, \tag{A.503}$$

$$\det(\mathtt{J}) = (\mathbf{r}_3^\top \mathbf{x}^h)^{-3}. \tag{504a}$$

$$\boldsymbol{\theta}^* = \underset{\boldsymbol{\theta}}{\mathrm{argmin}}\, J(\boldsymbol{\theta}) = \underset{\boldsymbol{\theta}}{\mathrm{argmin}}\, \left(-G(\boldsymbol{\theta}) + \lambda R(\boldsymbol{\theta})\right). \tag{505}$$

$$\begin{pmatrix} \mathbf{x}'_k \\ 1 \end{pmatrix} \sim \begin{pmatrix} \mathtt{R}(t_k\omega_Z) & t_k\mathbf{v} \\ \mathbf{0}^\top & 1 \end{pmatrix}^{-1} \begin{pmatrix} \mathbf{x}_k \\ 1 \end{pmatrix}, \tag{2.506}$$

$$\begin{pmatrix} \mathtt{A} & \mathbf{b} \\ \mathbf{0}^\top & 1 \end{pmatrix}^{-1} = \begin{pmatrix} \mathtt{A}^{-1} & -\mathtt{A}^{-1}\mathbf{b} \\ \mathbf{0}^\top & 1 \end{pmatrix} \tag{A.507}$$

$$\begin{pmatrix}\mathbf{x}'_k\\1\end{pmatrix}\sim\begin{pmatrix}\mathtt{R}(-t_k\omega_Z) & -\mathtt{R}(-t_k\omega_Z)(t_k\mathbf{v})\\\mathbf{0}^\top & 1\end{pmatrix}\begin{pmatrix}\mathbf{x}_k\\1\end{pmatrix}. \tag{508a}$$

$$\mathbf{x}'_k=\mathtt{R}(-t_k\omega_Z)(\mathbf{x}_k-t_k\mathbf{v}). \tag{509}$$

$$\mathtt{J}_k=\frac{\partial\mathbf{x}'_k}{\partial\mathbf{x}_k}=\mathtt{R}(-t_k\omega_Z), \tag{6.510}$$

$$\nabla\cdot\mathbf{f}=-2\omega_Z\sin(t\omega_Z). \tag{A.511}$$

$$\mathbf{x}_k^{h\prime}\sim\mathtt{R}(t_k\boldsymbol{\omega})\,\mathbf{x}_k^h, \tag{512a}$$

$$\det(\mathtt{J}_k)=(\mathbf{r}_{3,k}^\top\mathbf{x}_k^h)^{-3}. \tag{513}$$

$$\mathbf{x}'_k=\mathbf{x}_k-t_k\left(\mathbf{v}+(h_z+1)\mathtt{R}(\phi)\mathbf{x}_k-\mathbf{x}_k\right) \tag{3.514}$$

$$\det(\mathtt{J}_k)=(1+t_k)^2-2(1+t_k)t_k(h_z+1)\cos\phi+t_k^2(h_z+1)^2. \tag{A.515}$$

$$\begin{pmatrix}\mathbf{x}'_k\\1\end{pmatrix}\sim\begin{pmatrix}\beta_k\mathtt{R}(t_k\omega_Z) & t_k\mathbf{v}\\\mathbf{0}^\top & 1\end{pmatrix}^{-1}\begin{pmatrix}\mathbf{x}_k\\1\end{pmatrix}. \tag{516a}$$

$$\begin{pmatrix}\mathbf{x}'_k\\1\end{pmatrix}\sim\begin{pmatrix}\beta_k^{-1}\mathtt{R}(-t_k\omega_Z) & -\beta_k^{-1}\mathtt{R}(-t_k\omega_Z)(t_k\mathbf{v})\\\mathbf{0}^\top & 1\end{pmatrix}\begin{pmatrix}\mathbf{x}_k\\1\end{pmatrix}. \tag{517}$$

$$\mathbf{x}'_k=\beta_k^{-1}\mathtt{R}(-t_k\omega_Z)(\mathbf{x}_k-t_k\mathbf{v}). \tag{7.518}$$

$$\det(\mathtt{J}_k)=\beta_k^{-2}=\frac{1}{(1+t_ks)^2}. \tag{A.519}$$

$$R_D(\mathcal{E},\boldsymbol{\theta})\doteq\frac{1}{N_e}\sum_{k=1}^{N_e}\nabla\cdot\mathbf{f}_k.\quad\text{(mean)} \tag{520a}$$

$$R_A(\mathcal{E},\boldsymbol{\theta})\doteq\frac{1}{N_e}\sum_{k=1}^{N_e}|\det(\mathtt{J}(e_k))|\quad\text{(mean)} \tag{521}$$

$$S_k=\sum_i^{\mathrm{C}}\sum_j^{\mathrm{C}}|R_{k,i,j}|^2, \tag{4.522}$$

$$s_p=[S_1,S_2,...,S_p], \tag{A.523}$$

$$g(K,s_p)=\frac{\sum_{k=1}^K S_k}{\sum_{k=1}^p S_k}. \tag{524a}$$

$$\frac{\partial u}{\partial t}=-u\cdot\nabla u-\nabla p+\frac{1}{Re}\Delta u+\sin(ny)\hat{x},\quad\nabla\cdot u=0,\quad\text{on }[0,2\pi]^2\times(0,\infty) \tag{525}$$

$$\begin{aligned} \partial_t u(x,t) + \partial_x \left(u^2(x,t)/2\right) &= \nu \partial_{xx} u(x,t), \quad t \in (0,1] \\ u(x,0) &= u_0(x), \qquad\quad x \in (0,1) \end{aligned} \tag{1.526}$$

$$\begin{aligned} -\nabla \cdot (a(x) \nabla u(x)) &= f(x), \quad x \in (0,1)^2 \\ u(x) &= 0, \qquad\quad x \in \partial(0,1)^2 \end{aligned} \tag{A.527}$$

$$\begin{aligned} \partial_t w(x,t) + u(x,t) \cdot \nabla w(x,t) &= \nu \Delta w(x,t) + f(x), \\ \nabla \cdot u(x,t) &= 0, \quad x \in (0,1)^2 \\ w(x,0) &= w_0(x), \quad t \in (0,T] \end{aligned} \tag{528a}$$

$$\mathcal{G}_\theta := \mathcal{Q} \circ (W_L + \mathcal{K}_L) \circ \cdots \circ \sigma \left(W_1 + \mathcal{K}_1\right) \circ \mathcal{P}, \tag{529}$$

$$\left(\mathcal{K} v_t\right)(x) = \mathcal{F}^{-1}\left(R \cdot T_K\left(\mathcal{F} v_t\right)\right)(x) \quad \forall x \in D, \tag{5.530}$$

$$FMf \circ \otimes_X^M = \otimes_Y^M \circ MFf. \tag{A.531}$$

$$\begin{aligned} & LFX \to MFX \to FMX \to FMY \\ = \;& LFX \to FLX \to FLY \to FMY \\ = \;& LFX \to LFY \to FLY \to FMY \\ = \;& LFX \to MFX \to MFY \to FMY \end{aligned} \tag{532a}$$

$$(\phi' \circ \phi)(\psi' \circ \psi) = (\phi'\psi') \circ (\phi\psi) : FG \to F''G''. \tag{533}$$

$$\phi\psi : FH \to GK = \phi K \circ F\psi = G\psi \circ \phi H \tag{2.534}$$

$$\otimes_X^L\left(\bar{f}\right)(\bar{a}) = \prod_{i=1}^{n} f_i(a_i) \tag{A.535}$$

$$MX = \{m \in FX \mid \forall a \in X \; m(a) \in \mathbb{N}\}. \tag{536a}$$

$$DX = \{d \in FX \mid \sum_{a \in X} d(a) = 1\}. \tag{537}$$

$$X^{(n)} = \{m \in MX \mid \sum_{a \in X} m(a) = n\}, \tag{6.538}$$

$$\begin{aligned} & LLF \to MMF \to MFM \to FMM \to FM \\ = \;& LLF \to MMF \to MF \to FM \end{aligned} \tag{A.539}$$

$$\otimes_Y^L (LFh(\bar{f}))(\bar{b}) = FLh(\otimes_X^L(\bar{f}))(\bar{b}). \tag{540a}$$

$$\otimes_Y^L \circ LFh = FLh \circ \otimes_X^L. \tag{541}$$

$$F\mu_X^L \circ \otimes_{LX}^L \circ L\otimes_X^L = \otimes_X^L \circ \mu_{FX}^L. \tag{3.542}$$

$$\mu_{LX}^F \circ F \otimes_X^L \circ \otimes_{FX}^L = \otimes_X^L \circ L\mu_X^F. \tag{A.543}$$

$$\sum_{i:R=1} w_i \phi(X_i) = \frac{1}{N_{pop}} \sum_{j=1}^{N_{pop}} \phi(X_j) \tag{544a}$$

$$k(X_i, X_j) = exp(-||X_i - X_j||^2/b) \tag{545}$$

$$\min_w \; D(w,q) \quad \text{s.t.} \sum_{i:R_i=1} w_i \phi(X_i) = T, \; \sum_{i:R_i=1} w_i = 1, \text{ and } \; 0 \le w_i \le 1 \tag{7.546}$$

$$\sum_{i:R=1} w_i K_i = \frac{1}{N_{pop}} \sum_{j=1}^{N_{pop}} K_j, \; s.t. \sum_i w_i = 1, \; w_i \ge 0, \; \forall i \tag{A.547}$$

$$\sum_{i:R=1} w_i K_i = \sum_{j=1}^{N_{pop}} w_j^{(pop)} K_j, \; s.t. \sum_i w_i = 1, \; w_i \ge 0, \; \forall i \tag{548a}$$

$$\sqrt{\gamma} ||(w^{(pop)\top} \mathbf{V}_{pop} - w_s^\top \mathbf{V}_s) \mathbf{A}^{1/2}||_2 \tag{549}$$

$$\sum_{x \in X} \phi(x) w(x) p(X = x \mid R = 1) = \sum_{x \in X} \phi(x) p(X = x) \tag{4.550}$$

$$w(x) = \frac{p(x)}{p(x \mid R = 1)} = \frac{p(x)p(R=1)}{p(R=1|x)p(x)} = \frac{p(R=1)}{p(R=1|x)} \tag{A.551}$$

$$w_i = \frac{N_s/N_{pop}}{g(\phi(X_i)^\top \theta + \eta_i)} \tag{552a}$$

$$w(x) = \frac{p(x \mid R = 0)}{p(x \mid R = 1)} = \frac{p(R=0|x)p(R=1)}{p(R=1|x)p(R=0)} = \frac{p(R=0|x)}{p(R=1|x)} \frac{N_s}{N_{pop}} \tag{553}$$

$$\mu_s(w) = \frac{1}{N_s} \sum Y_i = \frac{1}{N_s} \sum w_i \phi(\mathbf{X})_i^\top \beta + \frac{1}{N_s} \sum w_i Z_i + \frac{1}{N_s} \sum w_i \nu_i \tag{1.554}$$

$$\mu = \overline{\phi(X)}_{pop}^\top \beta + \frac{1}{N_{pop}} \sum Z_i + \nu_i \tag{A.555}$$

$$\sum_{x \in X} \phi(x) w(x) p(X = x \mid R = 1) = \sum_{x \in X} \phi(x) p(X = x \mid R = 0) \tag{556a}$$

$$\begin{aligned} p(S=1) = logit^{-1}\Big(& PID(3way) + Age(4way) + Gender + educ(3way) + Race(4way) \\ & + BornAgain + PID(3way) * Age(4way) + BornAgain * Age(4way) \Big) \end{aligned} \tag{557}$$

$$\begin{aligned} p(Vote = D) = & PID(3way) + Age(4way) + Gender + educ(3way) + Race(4way) \\ & + BornAgain + PID(3way) * Age(4way) + BornAgain * Age(4way) \end{aligned} \tag{5.558}$$

$$\hat{\mu} = \sum_{i:R_i=1} w_i Y_i \tag{A.559}$$

$$Y = \phi(X)^\top \beta + Z + \nu \tag{560a}$$

$$w_i = h(\phi(X_i)^\top \theta + \eta_i) = \frac{1 - g(\phi(X_i)^\top \theta + \eta_i)}{g(\phi(X_i)^\top \theta + \eta_i)} \frac{N_s}{N_{pop}} = e^{-(\phi(X_i)^\top \theta + \eta_i)} \frac{N_s}{N_{pop}} \tag{561}$$

$$Y_i = \sum_{j=1}^{p} \beta_j X_{ij} + \epsilon_i, \quad \epsilon_i \sim N(0, \sigma^2), \tag{2.562}$$

$$\begin{aligned} &\beta_j \sim (1 - \gamma_j) N(0, \tau_j^2) + \gamma_j N(0, c_j^2 \tau_j^2); \\ &\gamma_j \sim Bern(\pi_j), \end{aligned} \tag{A.563}$$

$$\begin{aligned} &Y_i = \sum_{j=1}^{p} \beta_j \gamma_j X_{ij} + \epsilon_i, \epsilon_i \sim N(0, \sigma^2); \\ &\beta_j \sim N(0, s^2), \qquad \gamma_j \sim Bern(\pi_j). \end{aligned} \tag{564a}$$

$$\begin{aligned} \beta_{gj} = \gamma_g^{(1)} \gamma_{gj}^{(2)} b_{gj}, \quad \gamma_g^{(1)} \sim Bern(\pi_1), \quad \gamma_{gj}^{(2)} \sim Bern(\pi_2), \\ b_{gj} \sim N(0, s^2), \end{aligned} \tag{565}$$

$$\mathbf{Y} = \mathbf{XB} + \boldsymbol{\epsilon}, \quad \boldsymbol{\epsilon} \sim MN_{n \times q}(\mathbf{0}, \mathbf{I}_n \otimes \Sigma), \tag{6.566}$$

$$\begin{aligned} \gamma_{gj} \sim \text{Bern}(\pi_{gj}^{(\gamma)}), \quad \Phi^{-1}(\pi_{gj}^{(\gamma)}) = d_0 + d_1 A_{gj}, \quad d_0 \sim N(0, \sigma^2), \\ d_1 \sim N(\mu_d, \tau^2), \end{aligned} \tag{A.567}$$

$$\beta_{gj,k} | \mathbf{X}, \mathbf{Y} \sim l_{gj,k} N((1 - D_n) \hat{\beta}_{gj,k}^{LS}, \frac{1 - D_n}{n} \Sigma_{kk}) + (1 - l_{gj,k}) \delta_0(\beta_{gj,k}), \tag{568a}$$

$$BFDR(t) = \frac{\sum_{g=1}^{G} \sum_{j=1}^{m_g} P_{gj}(H_0 | \mathbf{X}, \mathbf{Y}) d_{gj}(t)}{\sum_{g=1}^{G} \sum_{j=1}^{m_g} d_{gj}(t)}. \tag{569}$$

$$s^2 | rest \sim IG\left(1 + \frac{pq}{2}, \nu + \frac{1}{2} Tr(\mathbf{B} \Sigma^{-1} \mathbf{B}^T)\right) \tag{3.570}$$

$$\alpha_g | rest \sim \text{Bern}(\pi_*^{(\alpha)}), \tag{A.571}$$

$$\gamma_{gj} | rest \sim \text{Bern}(\pi_{g*}^{(\gamma)}), \tag{572a}$$

$$\omega_{gj,k} | rest \sim \text{Bern}(\pi_{gj*}^{(\omega)}), \tag{573}$$

$$\Sigma | rest \sim \text{IW}\left(p + q + n, \mathbf{I} + (\mathbf{Y} - \mathbf{XB})^T (\mathbf{Y} - \mathbf{XB}) + s^2 \mathbf{B}^T \mathbf{B}\right) \tag{7.574}$$

$$Z_{gj}^{\mathcal{M}_S} = \frac{PIP_{gj}^{\mathcal{M}_S} - \frac{1}{R} \sum_{r=1}^{R} PIP_{gj}^{\mathcal{M}_S, r}}{\sqrt{\frac{1}{R} \sum_{r=1}^{R} (PIP_{gj}^{\mathcal{M}_S, r} - \left(\frac{1}{R} \sum_{r=1}^{R} PIP_{gj}^{\mathcal{M}_S, r}\right))^2}}, \tag{A.575}$$

$$(d_0, d_1) | rest \sim N\left((\Sigma_d^{-1} + \mathbf{A}^T \mathbf{A})^{-1} (\Sigma_d \vec{M}_d + \mathbf{A}^T \vec{\xi}), (\Sigma_d^{-1} + \mathbf{A}^T \mathbf{A})^{-1}\right) \tag{576a}$$

$$\nu^{(m)} = \frac{1}{\mathrm{E}_{\nu^{(m-1)}}\left[\frac{1}{s^2}|\mathbf{Y}\right]}, \tag{577}$$

$$\vec{b}_{gj}|rest \sim N(\vec{M}_{gj}, \Sigma_{gj}), \tag{4.578}$$

$$b_{gj,k}|\mathbf{X}, \mathbf{Y} \sim N(M_{gj,k}, \Sigma_{gj,kk}). \tag{A.579}$$

$$\lim_{n\to\infty} \frac{(1+ns^2)^{1/2}}{\exp\left(-\frac{1-D_n}{2\Sigma_{kk}}n\right)} = \lim_{n\to\infty} \frac{(s^2/2)(1+ns^2)^{-1/2}}{(-\frac{1-D_n}{2\Sigma_{kk}})\exp\left(-\frac{1-D_n}{2\Sigma_{kk}}n\right)} = 0, \tag{580a}$$

$$m(u,v) = \frac{1}{p}\mathbb{E}\left[\langle \mathbf{X}(u), \mathbf{X}(v)\rangle\right] = \alpha(u,v), \quad d(u,v) = \frac{1}{p}\mathbb{E}\left[\|\mathbf{X}(u) - \mathbf{X}(v)\|^2\right]. \tag{581}$$

$$\max_{i,j\in[n], i\neq j} |m(z_i, z_j) - \hat{m}(i,j)| \leq \max_{i,j\in[n], i\neq j} |\alpha(z_i, z_j) - \hat{\alpha}(i,j)|. \tag{1.582}$$

$$\max_{i,j\in[n], i\neq j} |\alpha(Z_i, Z_j) - \hat{\alpha}_{\mathrm{data}}(i,j)| \in O_{\mathbb{P}}\left(\frac{n^{2/q}}{\sqrt{p}}\right). \tag{A.583}$$

$$\max_{i,j\in[n], i\neq j} |\alpha(Z_i, Z_j) - \hat{\alpha}_{\mathrm{pca}}(i,j)| \in O_{\mathbb{P}}\left(\sqrt{\frac{nr}{p}} + \sqrt{\frac{r}{n}}\right). \tag{584a}$$

$$\max_{i,j\in[n], i\neq j} |m(Z_i, Z_j) - \hat{m}(i,j)| \tag{585}$$

$$\hat{\alpha}(u,v) = \frac{1}{|u||v|}\sum_{i\in u, j\in v} \hat{\alpha}(i,j), \quad \text{for all distinct pairs } u, v \in P_m. \tag{5.586}$$

$$m(z_i, z_j) - \hat{m}(i,j) \geq -M. \tag{A.587}$$

$$\hat{\alpha}(k,l) \geq H - M, \quad \forall k, l \in S_v. \tag{588a}$$

$$m(z_i, z_j) - \hat{m}(i,j) \leq M. \tag{589}$$

$$\max_{k\neq l\in[n]} \hat{\alpha}(k,l) \geq \alpha(z_i, z_j) - M, \tag{2.590}$$

$$\hat{\alpha}(k,l) \leq H - b + M, \quad \forall\, k \in S_v, l \in \bar{S}_v. \tag{A.591}$$

$$\max_{i\in[n]} \left\| p^{-1/2}\mathbf{Q}\zeta_i - \phi(Z_i)\right\| \in O_{\mathbb{P}}\left(\sqrt{\frac{nr}{p}} + \sqrt{\frac{r}{n}}\right). \tag{592a}$$

$$\mathbb{P}\left(\left|p^{-1}\left\langle \mathbf{Y}_i, \mathbf{Y}_j\right\rangle - \alpha(Z_i, Z_j)\right| \geq \delta \middle| Z_1, \ldots, Z_n\right) \leq \frac{1}{\delta^q} C(q,\varphi)\frac{1}{p^{q/2}} M(q, \mathbf{X}, \mathbf{E}, \mathbf{S}) \tag{593}$$

$$\mathbf{Y}_i = \gamma_i \mathbf{X}(i), \tag{6.594}$$

$$\left|p^{-1/2}\|\mathbf{X}(i)\| - 1\right| \leq \left|p^{-1}\|\mathbf{X}(i)\|^2 - 1\right| = \left|p^{-1}\sum_{j=1}^{p}\Delta_j(i)\right|, \tag{A.595}$$

$$\begin{aligned}\mathbb{E}[X_j(u)X_j(v)] &= \mathbb{E}\left[\mathbb{E}[X_j(u)X_j(v)|X_j(w)]\right] \\ &= \mathbb{E}\left[\mathbb{E}[X_j(u)|X_j(w)]\mathbb{E}[X_j(v)|X_j(w)]\right] \\ &= \mathbb{E}[X_j(w)^2].\end{aligned} \tag{596a}$$

$$m(u,v) = h(w) = \alpha(w,w) = \frac{1}{p}\sum_{j=1}^{p}\mathbb{E}[X_j(w)^2] = \frac{1}{p}\sum_{j=1}^{p}\mathbb{E}[X_j(u)X_j(v)] = \alpha(u,v), \tag{597}$$

$$\begin{aligned}\hat{h}(w) &= \hat{\alpha}(A',B') \\ &= \frac{1}{|A'||B'|}\sum_{k\in A',l\in B'}\hat{\alpha}(k,l) < \hat{H} = \hat{h}(u),\end{aligned} \tag{3.598}$$

$$\text{return}_i = \frac{p_i^{cl} - p_i^{op}}{p_i^{op}}, \tag{A.599}$$

$$\max_{i,j\in[n],i\neq j}|\alpha(z_i,z_j) - \hat{\alpha}(i,j)| < b/2, \tag{600a}$$

$$\hat{\alpha}(A,B) = \frac{1}{|A||B|}\sum_{k\in A,l\in B}\hat{\alpha}(k,l) \geq H - M. \tag{601}$$

$$p^{-1}\left\langle \mathbf{Y}_i, \mathbf{Y}_j\right\rangle - \alpha(Z_i,Z_j) = \sum_{k=1}^{4}\Delta_k \tag{7.602}$$

$$p(x_j) = \prod_{v\in\mathcal{V}} p\left(x_j(v)|x_j(\text{Pa}_v)\right), \tag{A.603}$$

$$\mathbb{E}\left[X_j(v)|X_j(\text{Pa}_v)\right] = X_j(\text{Pa}_v), \tag{604a}$$

$$\hat{\alpha}(w,w') = \frac{1}{|w||w'|}\sum_{i\in w,j\in w'}\hat{\alpha}(i,j). \tag{605}$$

$$\mathbb{E}[\|\mathbf{S}(Z_i)\mathbf{E}_i\|^2|Z_1,\ldots,Z_n] \leq \|\mathbf{S}(Z_i)\|_{\text{op}}^2\mathbb{E}[\|\mathbf{E}_i\|^2] \leq \max_{v\in\mathcal{Z}}\|\mathbf{S}(v)\|_{\text{op}}^2\, p\,\mathbb{E}[|\mathbf{E}_{11}|^2], \tag{4.606}$$

$$\|\mathbf{Y}_i - \mathbf{Y}_j\|^2 = \|\mathbf{Y}_i\|^2 + \|\mathbf{Y}_j\|^2 - 2\langle\mathbf{Y}_i,\mathbf{Y}_j\rangle, \tag{A.607}$$

$$\frac{1}{p}\mathbb{E}[\|\mathbf{Y}_i\|^2|\gamma_i] = \gamma_i^2\frac{1}{p}\mathbb{E}[\|\mathbf{X}(i)\|^2] = \gamma_i^2 h(i) = \gamma_i^2, \tag{608a}$$

$$\max_{i,j\in[n],i\neq j}|\alpha(i,j) - \hat{\alpha}_{\cos}(i,j)| \in O_{\mathbb{P}}\left(\frac{n^{2/q}}{\sqrt{p}}\right). \tag{609}$$

$$\max_{i\in[n]}\left|p^{-1}\|\mathbf{X}(i)\|^2 - 1\right| \in O_{\mathbb{P}}(n^{1/q}/\sqrt{p}). \tag{1.610}$$